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Evaluating Sentinel-2 gap filling techniques for cloud removal and data reconstruction

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Cloud cover creates frequent data gaps in high-resolution satellite imagery, particularly from Sentinel-2. These disrupt its continuity and reliability for time-sensitive applications such as water resource management, irrigation scheduling and crop health or yield prediction. Several approaches have been proposed in the literature, but there is a lack of performance comparisons. To address this challenge, this study aims to establish an evaluation framework to investigate the effectiveness of spatial, temporal, spatio-temporal, and spatio-spectral gap-filling approaches for restoring cloud-induced gaps in Sentinel-2 imagery using only satellite derived data. The evaluation was conducted using simulated cloud scenarios for both spatial within a single image and temporal for a time series cube of images. The methods were tested on key Sentinel-2 bands visible (B02, B03, B04), near-infrared (B08), and shortwave infrared (B11 and B12). The performance of each method was assessed using the coefficient of determination (R^2), rRMSE and bias. Among the evaluated methods, spatio-temporal category, especially clustering and linear regression (CLR), showed the highest accuracy and robustness across all gap scenarios and types. Spatio-temporal Deep learning (DL) also performed well but required more training effort and failed in generalizing over all scenarios. Also, spatio-spectral approaches like SSRF showed strong results in visible and NIR bands. In contrast, spatial methods such as kriging struggled with larger or irregular gaps. These findings offer a comparative evaluation of gap-filling methods, highlighting the strengths and limitations of each approach to guide the selection of appropriate methods and explore new prospects for improving gap-filling techniques. To ensure reproducibility all codes used in this study are publicly available on https://github.com/said-grich/gap-filling.

High resolution multispectral remote sensing data is a critical asset in various domains including precision agriculture, environmental monitoring, disaster management, forestry, water resource management and climate change studies. Several high resolution multispectral remote sensing products are available, each offering unique features and capabilities. For instance, the Landsat series, with its long-term archive, provides 30-meter resolution imagery across seven spectral bands. Landsat 8 and 9 continue to deliver today vital data for land use and environmental monitoring with a 8-day temporal resolution[1]. The Sentinel-2 mission with its twin satellites Sentinel-2 A and Sentinel-2B offers ten meters resolution for visible and near-infrared bands and twenty meters resolution for shortwave infrared bands with a temporal resolution of five days[2]. In addition, commercial satellites such as Worldview and GeoEye provide even higher spatial resolutions down to thirty centimeters allowing for extremely detailed analysis of small-scale features[3]. Each of these products serves specific needs, from broad-scale applications like environmental monitoring and climate change to fine-scale uses such as water resource management and precision agriculture. Sentinel-2 multispectral imagery offers balance of spatial, spectral, and temporal resolutions. This balance makes this mission a widespread data source for a wide range of agricultural applications, including irrigation and evapotranspiration estimation[4,5], yield prediction and land use land cover mapping[6–8], estimation of gross primary productivity and calculation of water use efficiency[9,10].

Despite the advantages of high-resolution optical satellite data, it faces significant challenges. The first and more frequent challenge is cloud cover. Optical satellites such as Sentinel-2 and Landsat cannot penetrate clouds because they block the transmission of radiation from the sun to the earth surface[11,12]. The second, and less evident, is sensor malfunctions. A notable example is the Scan Line Corrector (SLC) failure on the Landsat 7

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satellite in 2003, which led to substantial data gaps in the collected imagery. This results in substantial data gaps in the measured satellite images, which limit the utility of the data for continuous monitoring and precise analysis[13]. Many regions around the world, especially those with frequent cloud cover, can have significantly reduced data availability due to the limited number of cloud-free images. For example, accurate evapotranspiration estimation and irrigation planning require continuous data to track changes over time and respond promptly to crop water needs[14–16]. Similarly, yield estimation and land use mapping rely on consistent and complete datasets to produce accurate and reliable results[17]. Moreover, the limited availability of cloud-free images can delay the timely detection of crop stress and other critical conditions, affecting decision-making processes in precision agriculture.

To address these challenges, the remote sensing community has developed a variety of gap filling techniques that have evolved significantly over time. Early efforts focused on simple techniques such as kriging and inverse distance weighting (IDW), which utilized spatial correlations to estimate missing values[18,19]. As the complexity of challenges grew, more sophisticated methods emerged, including spatiotemporal models and advanced machine learning algorithms[20,21]. These improved both the accuracy and efficiency of gap filling by leveraging spatial and temporal information. In recent years, the integration of deep learning approaches has marked a significant advancement, enabling the capture of complex patterns within data and offering highly accurate reconstructions of missing pixels. These developed methods have successfully retrieved the missing data and maintain data continuity with high accuracy, resulting in improved applications of remote sensing data.

The primary objective of this study is evaluating the performance of different gap filling methods for optical satellites data. Specifically, identifying methods that can effectively handle variety of spatial distribution of gaps within the image with various sizes, as well as methods that are robust across different sizes of temporal gaps, including both sequential and separated temporal gaps. In addition, methods operate only using the existing satellite data without the need for any additional auxiliary data sources. The tested methods include spatial methods such as ordinary kriging, inverse distance weighting (IDW), Nearest-Neighbor, linear interpolation, and bicubic interpolation. Temporal methods are the same as spatial ones but operate along the temporal axis, aiming to fill gaps in time series vectors. Spatio-temporal methods include clustering and linear regression (CLR), dynamic time warping segmentation (DTWS), and spatio-temporal deep learning (DL). Finally, spatio-spectral methods include the spatial-spectral random forests (SSRF).

The rest of the paper is organized as follows: Sect. 2 provides a review of existing gap filling techniques. Section 3 outlines the materials and methods used in this study, including the study area, data collection, and the implementation of gap filling techniques. Section 4 presents the results of the comparative analysis, highlighting the performance of each method across different scenarios. Section 5 discusses the implications of the findings, focusing on the strengths and limitations of the evaluated methods. Finally, Sect. 6 concludes the paper by summarizing the key contributions of the study and suggesting directions for future research.

Literature review

In addressing the challenge of gap filling and cloud removal in optical remote sensing data, a variety of methodologies have been developed. These methodologies are categorized into spatial, temporal, spatiotemporal, and spatio-spectral categories, each designed to tackle different cases and scenarios (Table 1).

The spatial gap-filling category methods are based on the fact that the value of a missing data point is correlated with the values of neighboring points[22,23]. Common spatial gap-filling methods include geostatistical interpolation techniques such as kriging and IDW[18,19,24]. Additionally, algorithmic approaches such as the neighborhood similar pixel interpolator (NSPI)[25] and Generalized NSPI (GNSPI)[26] are also used. Studies have shown that geostatistical methods like kriging are highly effective for filling small spatial gaps due to their ability to model spatial dependencies accurately. However, these methods are computationally intensive and can be slow, making them less practical for large datasets or real-time applications. In contrast, algorithmic approaches such as NSPI and GNSPI have demonstrated better computational efficiency while maintaining accuracy, though they still face challenges in handling extensive datasets efficiently.

On the other hand, the temporal gap-filling methods assume that the value of a missing data point can be estimated using the temporal sequence of observations[27]. The temporal gap-filling methods include approaches such as linear regression[28] and statistical interpolation[29] as well as advanced machine learning techniques that use the relationship between time series data points to predict missing values[30,31]. For the Multisource fusion methods, it integrates data from multiple satellite sources to improve gap-filling accuracy leveraging the strengths of different sources[32]. The existing studies indicate that statistical methods are effective for small temporal gaps, but their performance degrades with larger gaps. Linear regression and similar approaches provide a straightforward and computationally efficient means of gap filling but are limited by their simplicity. Advanced machine learning approaches, like random forest regression, offer higher accuracy by learning complex temporal patterns in the data. However, they require significant computational resources and extensive training data. Additionally, these methods often need ancillary data such as climate data or fusion with other satellites data[32] to predict the missing values accurately, which can be a limitation in some scenarios.

Spatio-temporal gap filling methods leverage the information available across both spatial and temporal dimensions to achieve more precise and reliable gap filling. Different methods have been used, such as methods based on trend estimation and residual interpolation using geostatic models[20], machine learning models like spatio-temporal random forests (STRF)[33] and deep learning like long short-term memory (LSTM) architecture[34]. These methods integrate spatial and temporal predictors to enhance the estimation process, and they have shown superior performance compared to spatial or temporal methods. However, they are also more computationally intensive.

Finally, the spatio-spectral gap-filling methods assume that the spectral bands of the data contain complementary information that can be used to estimate missing values[27], leveraging the correlations between

Category	Method type	Method name	Data	Mission (Product)	References
Spatial	Geostatistical interpolation	Kriging	Reflectances	Landsat 7 ETM+ (SLC Off)	18
	Geostatistical	Kriging	Reflectances	Landsat 7 ETM+ (SLC Off)	19
	Algorithm	NSPI	Reflectances	Landsat 7 ETM+ (SLC Off)	25
	Algorithm	GNSPI	Reflectances	Landsat 7 ETM+ (SLC Off)	26
	Algorithm	Differential Dynamic Search Distance Algorithm (DDSDA)	Land Surface Temperature (LST), Evapotranspiration (ET)	MODIS (MOD11A2&MOD16A2)	39
	Segmentation	Multi-Scale Segmentation	Reflectances	Landsat 7 ETM+ (SLC Off)	40
	Algorithm of interpolation, Machine learning	Weather Research and Forecast Model (WRF), 3D Ordinary Kriging, Machine Learning	Air temperature	Estimated Temperature	41
	Fusion	Area-to-Point Regression Kriging (ATPRK)	Reflectances	Landsat 7 ETM+	22
Temporal	Algorithm	Linear regression	LAI	MODIS (MCD15A2H)	28
	Algorithm	Multisource fusion	Reflectances	MODIS (MOD9A1) And Landsat (Landsat5, Landsat7)	32
	Machine Learning	Gaussian Process Regression (GPR)	LAI	Sentinel-2	42
	statistical	Inversed square distances (ISD), Arithmetic mean (AM)	rainfall data	Stations data	29
	Algorithm	discrete Fourier transform (DFT)	Weather Data	Stations data	43
	Machine learning	ClimateFiller	Weather Data	Era5, Stations data	30
	Machine learning	Random forest regression	Soil moisture	Global Navigation Satellite System (GNSS)	31
Spatio-Temporal	Algorithm	Generalized Additive Model (GAM)	Enhanced vegetation index (EVI)	MODIS (MOD13A1)	20
	Algorithm	mean distance-weighted ratios from multiple neighboring pixels time series.	Land Surface Temperature (LST), Bidirectional Reflectance Distribution Function (BRDF)	MODIS (MOD11A2, MCD43B4)	44
	Geostatistical	Spatio-Temporal Geostatistical Prediction	Atmospheric CO2	Greenhouse Gases Observing Satellite (GOSAT)	45
	Geostatistical	Kriging	Air quality indicator	AirBase2	46
	Algorithm of interpolation, Machine learning	K-means clustering, Neighborhood Similar Pixel Interpolator (NSPI)	Reflectances	Landsat 7 ETM+ (SLC Off)	35
	Algorithm	Quantile Regression	Normalized Difference Vegetation Index (NDVI)	MODIS (MOD13A1)	47
	Deep Learning	Long Short-Term Memory (LSTM)	temperature, specific conductance (SpC), water table elevation	Groundwater monitoring data	48
	Fusion	Spatial and Temporal Adaptive Gap-Filling Method (STAGFM)	Normalized Difference Snow Index (NDSI)	MODIS (MYD10A1, MYD10A1)	49
	Machine learning	K-means Clustering, LASSO regression	Land Surface Temperature (LST)		50
	Machine learning	SSGP-toolbox	Land Surface Temperature (LST), Surface Albedo, Normalized Difference Vegetation Index (NDVI)	Sentinel-3 (SLSTR), MODIS (MOD09GA, MOD11A1)	51
	Deep Learning	U-Net	Quasi-global carbon monoxide observations (NO2)	Sentinel-5P	52
	Deep Learning	Content-Sequence-Texture Generation (CSTG) network	Reflectances	MODIS (MOD09A1, MOD13Q1)	53
	Algorithm	Differential Dynamic Search Distance Algorithm (DDSDA)	Land Surface Temperature (LST), Evapotranspiration (ET)	MODIS (MOD11A2 & MOD16A2)	39
	Machine learning	Spatial-Temporal Random Forest (STRF)	Land Surface Temperature (LST)	Landsat 8, Geostationary Operational Environmental Satellites (GOES)	33
Spatio-Spectral	Algorithm	spatial-spectral radial basis function (SSRBF)	Reflectances	Landsat 7 ETM+	54
	Machine learning	spatial-spectral random forest (SSRF)	Reflectances	Landsat 8 (OLI), Sentinel-2 (MSI)	21

Table 1. Summary of existing gap-filling methods.

different spectral bands. Studies have demonstrated that spatio-spectral methods can significantly enhance gap filling accuracy. For instance, spatial-spectral random forests (SSRF)[21] have shown robust performance in filling gaps in Landsat 8 and Sentinel-2 data compared to methods such as based spectral random forest (SRF) and MNSPI[35].

Recent studies have highlighted the lack of comprehensive comparative studies focused on optical satellite data. A previous work[36] evaluated gap-filling approaches for satellite-based PM2.5 (pollution monitoring) prediction models, demonstrating that decision tree-based methods and multiple imputation (MI) strategies

effectively address data gaps. Similarly, the study[37] conducted a comparative study on various methods for producing all-weather daily remotely sensed near-surface air temperature, underscoring the critical role of spatial, temporal, spatiotemporal, and multisource fusion-based gap-filling techniques. Additionally[38], compared gap-filling techniques applied to the ESA CCI soil moisture dataset, highlighting the effectiveness of SVMs and the importance of dynamic and static variables in improving gap-filling accuracy. This underscores the significant need for comparative studies that highlight the advantages and limitations of gap-filling techniques applied to optical satellites in various scenarios, as well as the computational needs for each method.

For this study, a list of methods is selected to address the challenge of gap filling caused by cloud cover across various categories. This includes spatial category which is the geostatistical and interpolation methods like ordinary kriging, Inverse Distance Weighting (IDW), Nearest-Neighbor, linear, and spline interpolation. The category of temporal method is addressed with similar geostatistical and interpolation techniques but in temporal terms. For the spatiotemporal category, Clustering and Linear Regression Method based on machine learning techniques[50], DTWS region-growing segmentation and spatio-temporal deep learning via the U-net model architecture[52] are considered. The spatio-spectral category is tackled with the spatial-spectral random forests (SSRF) method[21]. Methods that rely on data fusion[22] or ancillary data sources are excluded to maintain the independence of the gap-filling process from external data inputs.

Materials and methods

Data collection and study area

The gap filling methods are evaluated over a heterogeneous study area located in the Tensift basin, Morocco (31.60° N, 7.98° W) (Fig. 1). The study area is characterized by a diverse land cover that includes urban areas, water bodies, vegetation cover mainly olive trees and bare soil. The climate of the Tensift basin is semi-arid Mediterranean, with average annual precipitation ranging from 150 mm in the lowlands to over 500 mm in the mountainous regions. The precipitation is highly variable, both spatially and temporally, with the majority occurring during the winter months[55].

The dataset under this study is obtained from the Sentinel-2 satellite mission for the period from April 1, 2022, to August 9, 2022. Sentinel-2 is a high-resolution optical imaging satellite mission operated by the European Space Agency (ESA) as part of the Copernicus program. It consists of two satellites, Sentinel-2 A and Sentinel-2B, which provide a five-day revisit time. The mission captures imagery across thirteen spectral bands with varying spatial resolutions: four bands at 10 m (B02 - blue, B03 - green, B04 - red, B08 - near-infrared), six bands at 20 m (B05, B06, B07 - red edge, B8A - narrow NIR, B11 - short-wave infrared 1, B12 - short-wave infrared 2), and three bands at 60 m (B01 - coastal aerosol, B09 - water vapor, B10 - cirrus). In order to properly evaluate the methods, the time period is chosen so that as many images as possible are cloud-free over the study area. Indeed, only three images are contaminated by clouds out of 27 images available during the study period (Fig. 2). Among the thirteen Sentinel-2 spectral bands, six bands (B02, B03, B04, B08, B11, B12) were selected for this study due to time requirements associated with filling gaps in each band. These selected bands are critical for many remote sensing applications. Moreover, they cover two key resolutions ten meters and twenty meters.

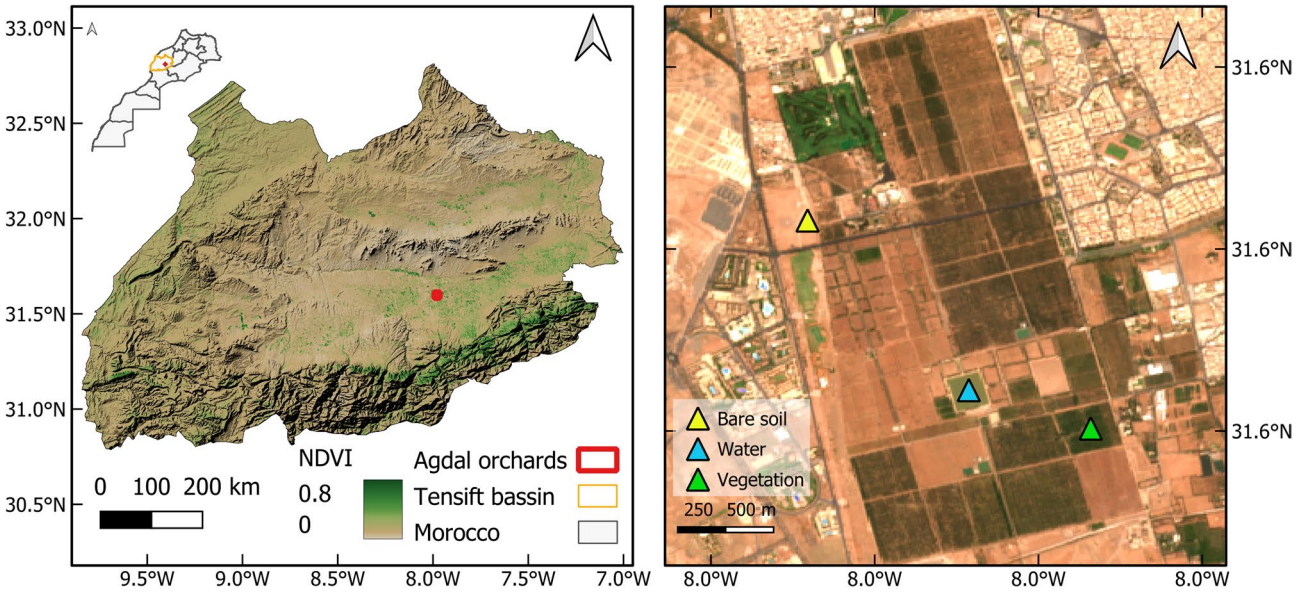


Fig. 1. Location of the "Agdal" study area (right) within the Tensift basin in Morocco (left). The maps display topography (SRTM DEM) and 2024 Sentinel-2 (S2) NDVI for the basin, and natural color RGB composite (S2 bands B4/B3/B2, 2024) for the orchard. Data processed via Google Earth Engine (GEE) and visualized using QGIS[56] (https://qgis.org/).

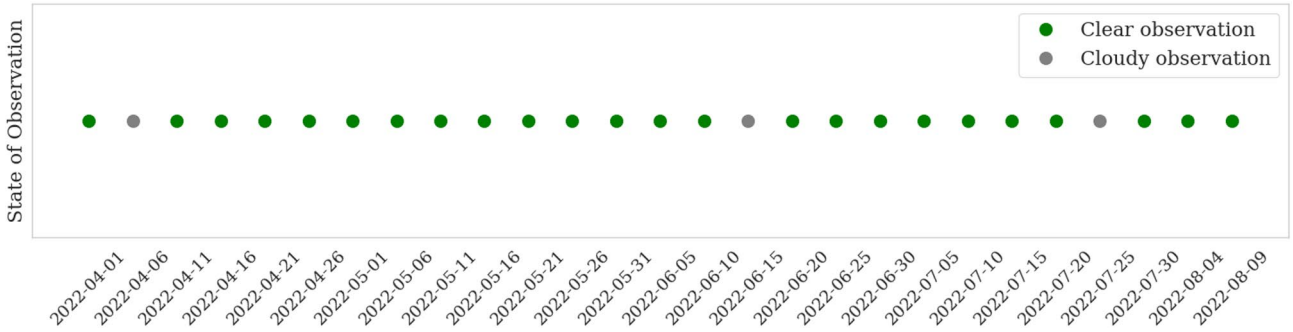


Fig. 2. Period of study with clear and cloudy Sentinel-2 images.

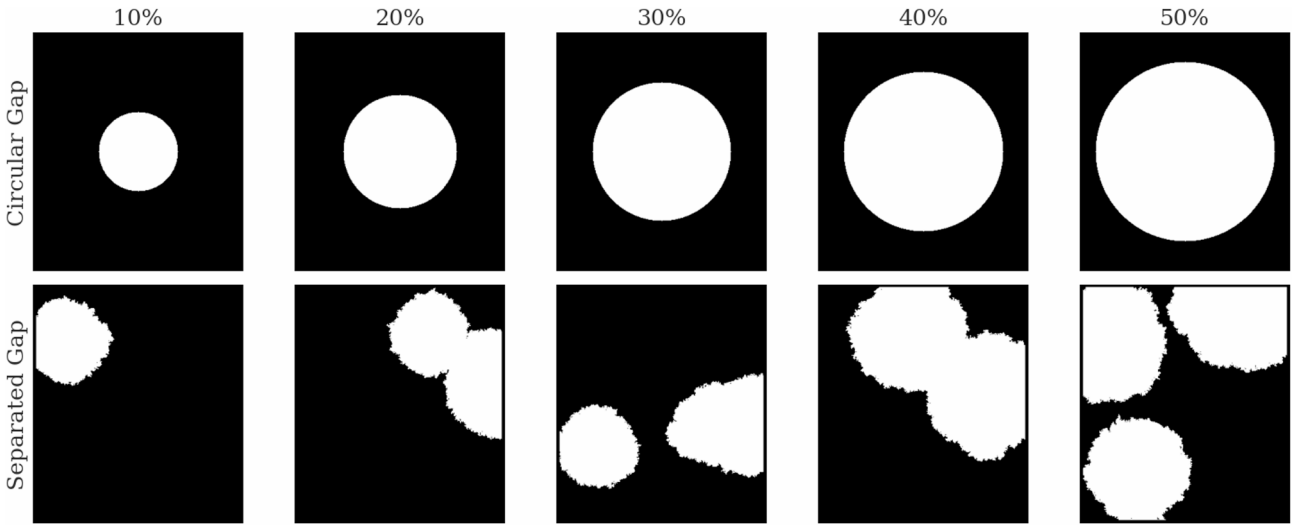


Fig. 3. Spatial Gaps masks for the two scenarios circular and separated and the five gap sizes for each scenario.

Simulated cloud-induced gaps

To evaluate the gap-filling methods, various scenarios of simulated gaps are investigated. Two types of gaps are introduced: (i) spatial gaps, where groups of adjacent pixels are removed within a single image, and (ii) temporal gaps, where a percentage of dates are removed. For the spatial gap, two scenarios with two spatial distributions are considered: circular gaps and separated gaps. Circular gaps are designed to simulate a single cloud patch within the center of the images. The size of this patch represents the percentage of missing pixels and ranges from 10% to 50% of the image's total pixel count. For the separated gaps, they aim to more accurately reflect the irregular spatial distribution of clouds within the image, unlike the uniformity of the circular gap. For both types, five gap sizes ranging from 10% to 50% are simulated. In particular, for the separated gaps case, a single cloud patch is generated for the 10% and 20% gaps, while two patches are adopted for the 30% and 40% gaps, and three patches for the 50% gap (Fig. 3). Similarly, two scenarios are distinguished for the temporal gap: sequential and separated gaps. Sequential gaps are characterized by continuous cloud-affected images within the center of the time series. In contrast, the cloudy images in the separated gap are dispersed randomly throughout the time series (Fig. 4).

Gap-filling techniques

Spatial methods

Spatial gap-filling methods exploit the spatial autocorrelation between neighboring pixels[26]. The primary objective is to estimate missing pixel values using information from surrounding pixels. Geostatistical methods like ordinary kriging and IDW predict unknown values by assigning weights to neighboring pixel values based on their distance or statistical relationships. Ordinary kriging models spatial autocorrelation through a variogram, which describes how variance changes with the distance between points[57]. It assigns optimal weights to neighboring pixels, balancing proximity and similarity, and provides both predictions and uncertainty estimates. This makes it effective for complex spatial patterns but computationally demanding. In contrast, IDW assigns higher weights to closer pixels based only on distance, without considering spatial structure[58]. While faster and easier to implement, it may struggle with accuracy in heterogeneous areas.



Fig. 4. Temporal Gaps masks for the two scenarios sequential and separated and the five gap sizes for each scenario.

Interpolation methods, including Nearest-Neighbor, linear, and cubic spline estimate missing values by fitting smooth functions between known pixel values[59]. Nearest-Neighbor simply assigns the value of the closest known pixel, while linear and cubic spline interpolations construct continuous functions that preserve smooth transitions.

Temporal methods

Similarly to the spatial features, the temporal feature has a pattern within time series data, presenting a source of information that can be used in gap filling. Consequently, all the aforementioned methods, such as ordinary kriging, IDW, Nearest-Neighbor, and linear interpolation, are tested using the temporal dynamics variation to reconstruct the missing values.

Spatiotemporal methods

Clustering and linear regression (CLR) method The Clustering and Linear Regression (CLR) method[50], originally developed for gap-filling in Landsat-derived Land Surface Temperature (LST) products, combines clustering analysis with regression modeling to exploit temporal similarity among pixels. The detailed procedure is summarized below:

1. Training data preparation: A dense dataset of clear observations is collected from orbit overlap regions, where multiple acquisitions increase temporal coverage.
2. Harmonic reconstruction: To handle cloud-contaminated records in the overlap regions, a 7-parameter harmonic model is applied to reconstruct seasonal trajectories, ensuring that the training dataset preserves realistic seasonal dynamics.
3. Clustering: The prepared time series are grouped into 20 clusters using the k-means algorithm, with each cluster centroid representing the mean seasonal profile of that group.
4. Distance calculation: For each target pixel with missing values, the Euclidean distance between its partial time series and each cluster centroid is computed.
5. Stratified sampling: Based on these distances, 100 pixels are randomly sampled from the most similar clusters using inverse-distance weights, producing a representative training dataset for the target pixel.
6. Regression modeling: A Lasso linear regression model is built using the sampled pixels as predictors to estimate the missing values of the target pixel.

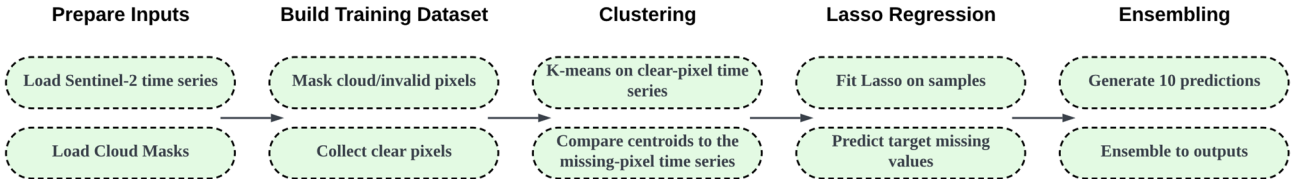


Fig. 5. Clustering and Linear Regression (CLR) method Workflow for Sentinel-2 Gap-Filling.

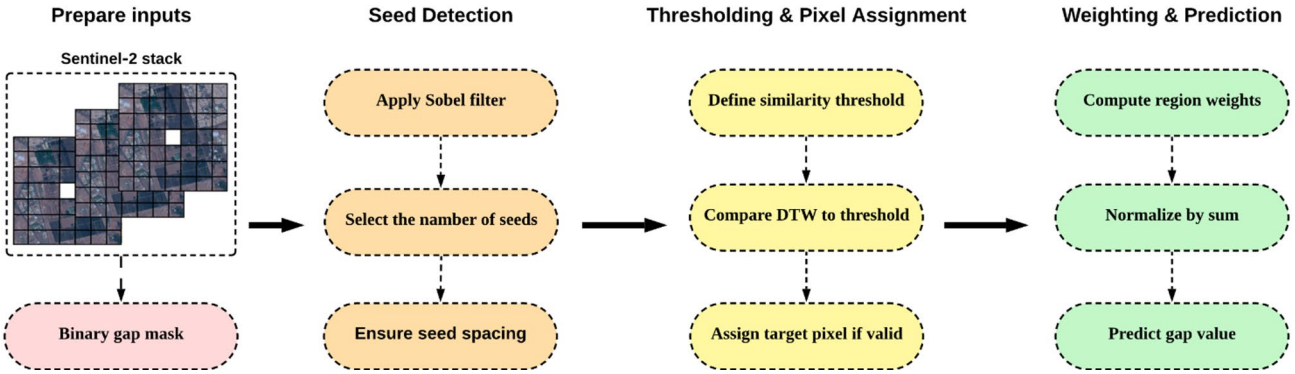


Fig. 6. Workflow of the DTWS Method for Sentinel-2 Gap Filling.

7. Ensemble prediction: To enhance robustness, the sampling and regression procedure is repeated 10 times, and the median of the predictions is used as the final gap-filled value.

8. Uncertainty estimation: The standard deviation across iterations is recorded and reported as an indicator of prediction uncertainty.

The CLR method is based on two key principles: the assumption that pixels with similar time series behavior can provide reliable estimates for each other, and the use of multiple linear regression modeling to predict missing values. First, a dense dataset is used for training, taking advantage of the high density of time series data in the overlapping tiles. Then, this dataset is clustered to gather similar pixels in the time using the Euclidean distance. Subsequently the clusters are stratified based on time series similarity to the target missing pixel. Finally, a linear regression model is created to predict missing values based on stratified training data in the time series of the target pixel.

In this work, the cloud-free areas are used instead of orbit overlapping tiles data, taking advantage of the temporal resolution of Sentinel-2 (five days). An overview of the adapted CLR workflow is shown in (Fig. 5). Firstly, k-means is used to categorize clear observation into distinct clusters[60]. Second, the Euclidean distance from the cluster averages to the target pixel is calculated. Using this distance inversely as weights, a stratified random sampling of clusters similar to the target pixel is obtained. Subsequently, 100 pixels are selected from the sampled clusters to train a Lasso linear model[61] that will be used to predict the missing values. For each target pixel, the selection and prediction processes are iteratively executed ten times[50], integrating ensemble learning principles using the median to enhance the robustness of the predictions.

Dynamic time warping segmentation (DTWS) method In the CLR method, the K-means clustering technique is used to identify the clusters based on seasonality patterns. Alike, the DTWS method implements spatio-temporal region-growing segmentation to identify similar regions. This segmentation is achieved using Dynamic Time Warping (DTWS) similarity to group similar pixels into same regions across both space and time dimensions[62]. Once these pixels are grouped, the pixel weighting technique is applied to accurately predict and restore missing pixels within the image[40]. The DTWS procedure can be summarized in the following steps (Fig. 6):

1. Seed initialization: Pixels are grouped into regions starting from seed points. To avoid edge effects, seed points are identified using the Sobel filter[63].

2. Seed balance: The number of seeds strongly affects the results. Too few seeds lead to under-segmentation (different regions merged), while too many causes over-segmentation. In this study, an optimal value of 100 seeds was used (Table 2).

3. Threshold definition: Each region is represented by the mean of its time series, which serves as the threshold reference (Table 2). Candidate pixels are compared to this mean using DTW distance.

4. Pixel assignment: If the DTW distance is below the threshold, the pixel is added to the region; if it is above the threshold, it is excluded.

5. Weight calculation: each region is summarized by the mean of its time series, which is then compared to the target (missing pixel). A weight is assigned to each region as the inverse of its DTW distance to the target (**Eq-1**).

Name of Parameter	Value
Number of Seeds	100
Distance Between Seeds	20
Threshold	Mean
Number of Regions for Estimation	10

Table 2. DTWS region-growing segmentation algorithm parameters.

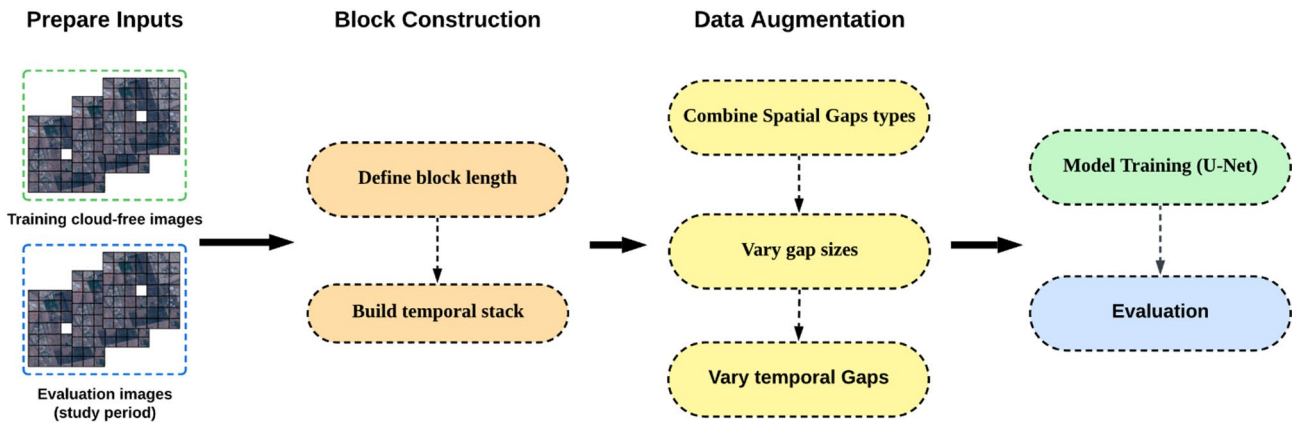


Fig. 7. Workflow of the DL Method for Sentinel-2 Gap Filling.

6. Similarity ranking: Regions are ranked by similarity, and the 10 most similar regions are selected (Table 2).
7. Prediction: The missing pixel value is estimated as a weighted average of the selected regions **(Eq-3)**, with weights normalized according to (**Eq-2**).

$$Wi = \frac{1}{DTW(A, Bi)} \tag{1}$$

$$Wi' = \frac{Wi}{\sum_{j=1}^{n} Wj} \tag{2}$$

$$Vgap = \sum_{i=0}^{n} (Vi \times Wi\prime) \tag{3}$$

Where $DTW(A, Bi)$ is the dynamic time warping distance between the missing pixel time series (A) and the i-th similar region time series (Bi), Vi is the mean of the region time series of the i-th similar region, n is the number of similar regions considered, Wi is weight of i-th region, Wi' is normalized weight of i-th region and $Vgap$ is predicted value of the missing pixel.

Spatiotemporal deep learning method (DL) The DL method is based on a U-Net architecture with partial convolutional layers. U-Net is a type of convolutional neural network (CNN)[64], originally designed for biomedical image segmentation. It is characterized by its U-shaped structure, which consists of an encoder (contracting path) and a decoder (expansive path). The encoder extracts features through successive convolutional and pooling layers, while the decoder reconstructs the image using transposed convolutions to restore spatial resolution, allowing for precise localization. This method was used by[65] for inpainting task to complete the damaged RGB images. Subsequently, the same architecture was used to fill the gaps in Sentinel-5 products[52]. In this study the DL method was employed to fill the gaps in Sentinel-2 multispectral imagery following successive steps (Fig. 7):

1. Mask generation: Binary masks are created, where zeros represent missing pixels (gaps) and ones represent valid pixels.
2. Block construction: The input to the DL model is a stack of Sentinel-2 images, hereafter called a block. Each block captures both the temporal variation (image sequence) and the spatial information (pixels within each image).
3. Training dataset preparation: A set of 176 cloud-free images from 2017 to 2021 was selected. Eight images were sampled every three months to form blocks that capture representative seasonal variations (Table 3).
4. Data augmentation: To build a robust training dataset, all combinations of cloud types, gap sizes, and temporal arrangements were generated. This produced 1,750 blocks, which were split into 80% for training and 20% for testing.
5. Model training and evaluation: The U-Net with partial convolutional layers was trained using the prepared dataset. The model was then evaluated during the same study period as the other gap-filling methods (Fig. 2).

Name of Parameter	Value
Image size	364
Block size	8
Batch size	16
Number of epochs	50
Number of partial convolutional blocks	2
Learning rate	0.001

Table 3. Selected hyperparameters of the U-Net model with partial convolutions.

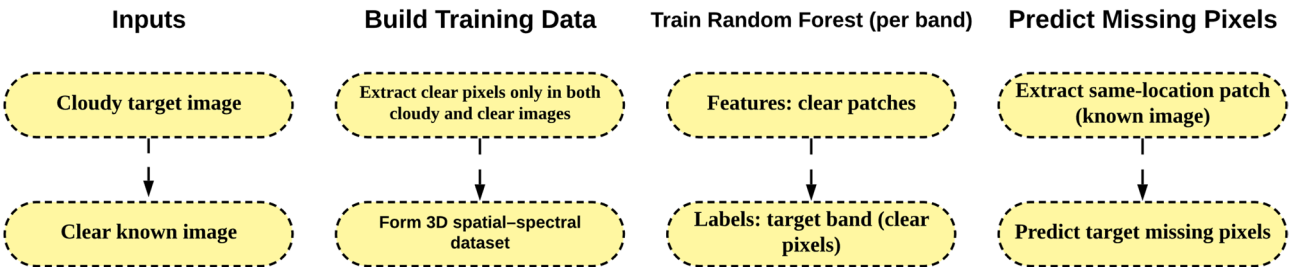


Fig. 8. Spatial–Spectral Random Forest (SSRF) Workflow for Gap-Filling Sentinel-2 Imagery.

1.1.1 Spatio-spectral random forest (SSRF)

The SSRF method was proposed to fill gaps in multispectral satellite data from Landsat 8 and Sentinel-2[21]. Based on the spatially adjacent and multispectral information from a known image, the approach aims to fill the gap in each band separately in a cloudy image based on the random forest algorithm (Fig. 8). The process begins with the identification and extraction of non-cloudy data from both the target band in the cloudy image and the known image. These pixels are transformed into a 3D spatial-spectral dataset with a patch size of 3×3 pixels, capturing spectral information across multiple bands and spatial information from neighboring pixels. This data is then used to construct training samples where the target band acts as the dependent variable for the random forest algorithm. This setup enables the SSRF to learn the relationships between different spectral bands and spatial context. Once the model is trained, the missing pixels are predicted by using their corresponding pixels in the known image.

Evaluation criteria

A comprehensive evaluation framework is established to assess the performance of the selected gap filling methods. For this evaluation, a specific period is selected excluding cloudy images (Fig. 2). The image of June 5, 2022 is considered as ground-truth observation to calculate the evaluation metrics. In order to assess the effectiveness of the spatial methods, the spatial gaps (circular and separated) with various sizes (from 10% to 50%) are considered (Fig. 3). This test aims to demonstrate the performance of these methods in handling both large and small, centered, and dispersed clouds (Fig. 9). To evaluate temporal methods, temporal gaps of different types and sizes, including sequential and separated gaps, are used (Fig. 4). To take into consideration the effect of land cover type on the gap filling temporal methods different cover types are considered for the evaluation such as (bare soil, vegetation, and water) (Fig. 1). Time series vectors for each pixel representing a specific cover type, representing the spectral changes over selected cloud-free periods (Fig. 9). Regarding the spatiotemporal methods, the evaluation framework incorporates varied test scenarios. Firstly, to assess spatial robustness, a dataset with circular and separated spatial gaps ranging from 10% to 50% with a 30% temporal separated gap is created. Secondly, to evaluate temporal robustness, sequential and separated temporal gaps varying between 10% and 50% are applied with a constant spatial separated gap of 30% (Fig. 9). Like spatial methods, the performance of spatio-spectral methods is evaluated across different spatial gap types and sizes. Since the spatio-spectral methods in this study incorporate temporal aspects by training the model using a known image, the period between the known image and the target cloudy image is varied to observe its impact on the performance. In addition, the combination of bands used in training is considered as a factor that may influence the results of the spatio-spectral methods. Therefore, different combinations of bands are selected in training to fill a single band (Fig. 9).

The best-performing method from each category is evaluated for the six selected Sentinel-2 bands using four statistical metrics: relative root mean squared error (rRMSE), root mean square error RMSE, coefficient of determination (R^2) and bias. These metrics are calculated using the filled part of the artificial gap, as well as the observation of June 5, 2022, used as the ground truth image. To evaluate the effect of gap size in case of spatial circular gap, a 10% gap is used as a baseline due to its common area with the other gap sizes. In contrast, for spatial separated gaps, no baseline area is available due to the varied locations of gaps for the different gap sizes.

For the visual evaluation, eight Sentinel-2 images are selected from the period between May 31, 2023, and July 5, 2023. The image dated June 25, 2023, is used as the target scene and is cloud-free. To create a realistic

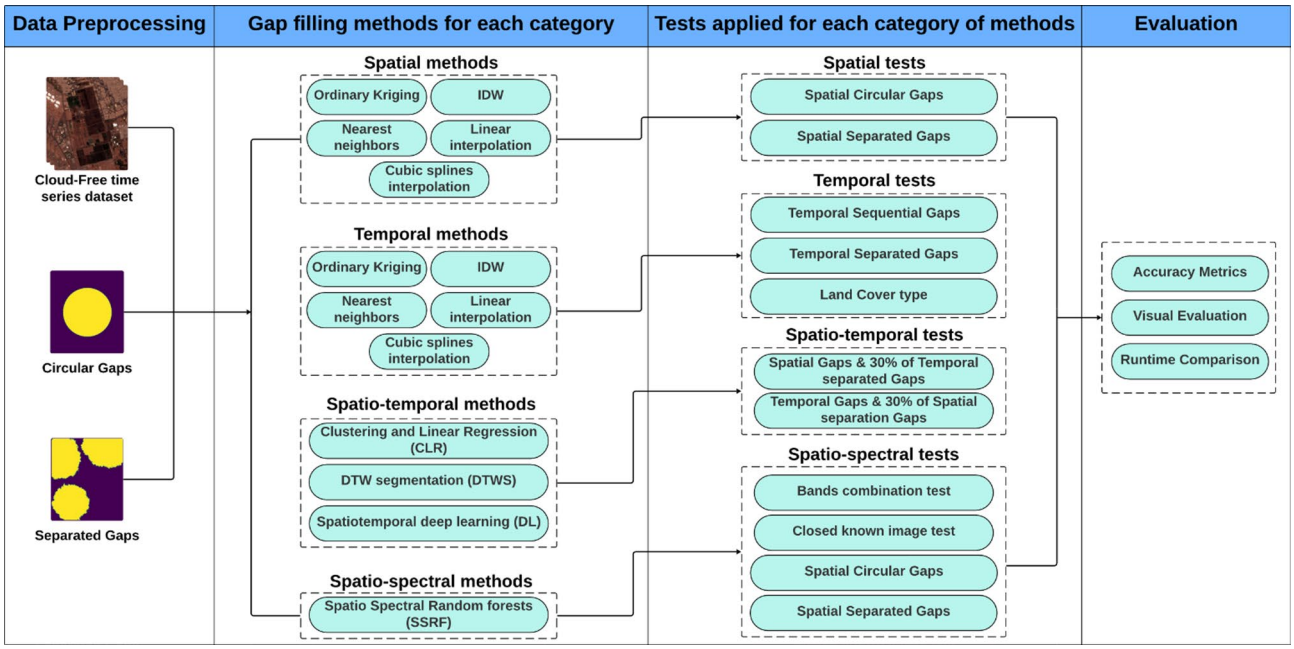


Fig. 9. Evaluation framework for Sentinel-2 gap filling methods.

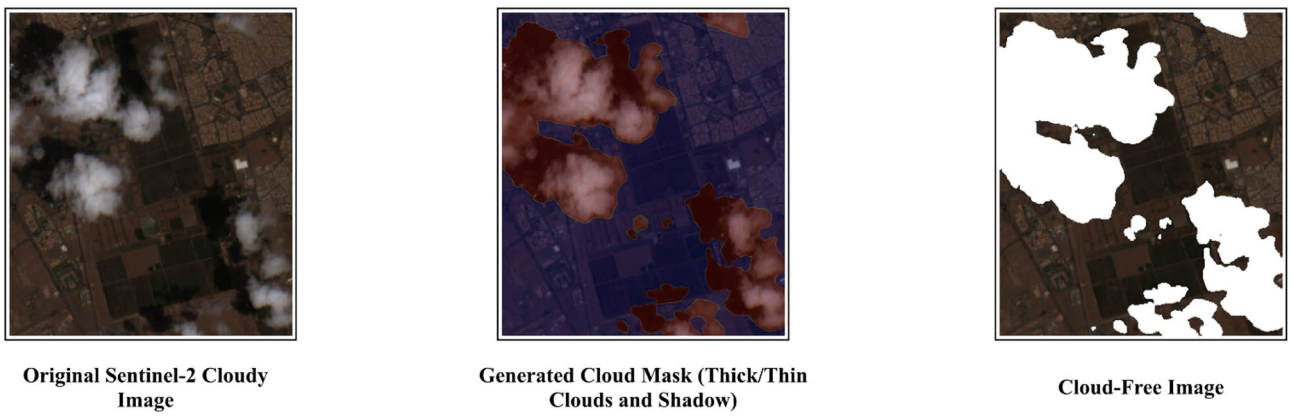


Fig. 10. Cloud Mask Generation Process for Gap-Filling in Sentinel-2 Imagery.

gap-filling problem, we apply a real cloud-and-shadow mask extracted from a different observation (June 20, 2023) onto this clear scene. The selected dataset represents a real use case of utilization of gap-filling techniques, providing a practical scenario to assess the effectiveness of different methods. The cloud mask extraction process is a crucial step in preparing the input for each image to be filled. In this study, the SEnSeI-v2 deep learning model[66] is used. This model is efficiently generalized to different sensors enabling accurate cloud and shadow masking on various satellite images. The cloud mask extraction is performed on all images in the visual test dataset to prepare the input for each gap filling method (Fig. 10). The extraction process ensures that all clouds and their associated shadows are accurately identified and masked. All RGB composites use the natural color configuration (B4/B3/B2) with a single global linear display stretch (reflectance clipped to [0, 0.3] and no gamma or local enhancement). As a quantitative evaluation, SSIM was computed against the original cloud-free June 25, 2023, scene, using only the masked pixels (the gap area) and the original reflectance values; identical settings were applied to all methods.

Results

This section presents the evaluation results of the gap filling methods including statistical metrics, visual evaluation and running time comparison. For each category, the methods are first compared for the band B04, and the best one is then chosen to evaluate its performance for the other bands.

Spatial methods

Figure 11; Table 4 summarize the statistical metrics obtained for the different scenarios tested for the spatial methods. The statistical analysis reveals suboptimal performance for spatial methods, particularly as the complexity and size of gaps increase. Ordinary kriging is the outperformed method with an rRMSE of 37.27% at 30% of circular gap and 31.48% for the same gap size in the separated gaps scenario (Fig. 11a). Nearest-Neighbor interpolation showed a higher rRMSE compared to the ordinary kriging, IDW and the bilinear interpolation for gap sizes up to 40% (60.80% for circular gap and 54.57% for separated gap at 40% of gap size). This indicates the limited ability of this method in handling large gaps (Table 4). Although IDW and bilinear are slightly more efficient than Nearest-Neighbor interpolation, they still show high rRMSE values compared to ordinary kriging, especially for small gap sizes (≤ 20%).

In the circular gap scenario, the ordinary kriging maintains a lower rRMSE that increases gradually from 24.84% at 10% gap size to 42.85% at 50% (Fig. 11b). In contrast, the separated gap scenario showed an irregular performance as the gap size is increasing. This inconsistency in the method performance for this scenario is related to the heterogeneity of cover types within the image. As the gap size increases, the location of the separated gaps in the image has changed to cover a new area not covered by the previous smaller size, which may include a different cover type.

The bicubic interpolation method showed the poorest performance among all spatial methods. For instance, in circular gaps it starts with an extremely high rRMSE of 146.18% for a 10% gap and reaches 4083.09% at a 50% gap size (Table 4). For separated gaps, the method shows a low performance with an rRMSE values higher than 109.39% in all sizes.

Similarly, the performance of ordinary kriging for the Sentinel-2 spectral bands at 10% spatial gaps indicates low effectiveness, with modest R^2 values across both circular and separated gap scenarios (Fig. 11c). For the visible bands (B02, B03, B04), the R^2 values remain low, ranging between 0.15 and 0.19 for circular gaps and 0.23–0.25 for separated gaps. The SWIR bands (B11, B12) show slightly better performance, with R^2 values of 0.11–0.30 for circular gaps, and 0.31–0.40 for separated gaps, respectively. Notably, band B08 exhibits the poorest performance, with an R^2 of only 0.05 for separated gaps, although it improves somewhat to 0.25 in the circular gap scenario. Despite this relative improvement, the overall correlation for B08 remains weak.

Temporal results

The evaluation of temporal gap filling methods in Sentinel-2 bands showed that ordinary kriging outperformed other methods, achieving low rRMSE values of 5.44% over vegetation cover, 6.20% over bare soil, and 2.90% over water bodies at 30% of sequential temporal gap (Fig. 12a). The IDW method performance is close to the ordinary kriging with rRMSE values of 4.90% over vegetation cover, 6.45% over bare soil, and 3.95% over water bodies

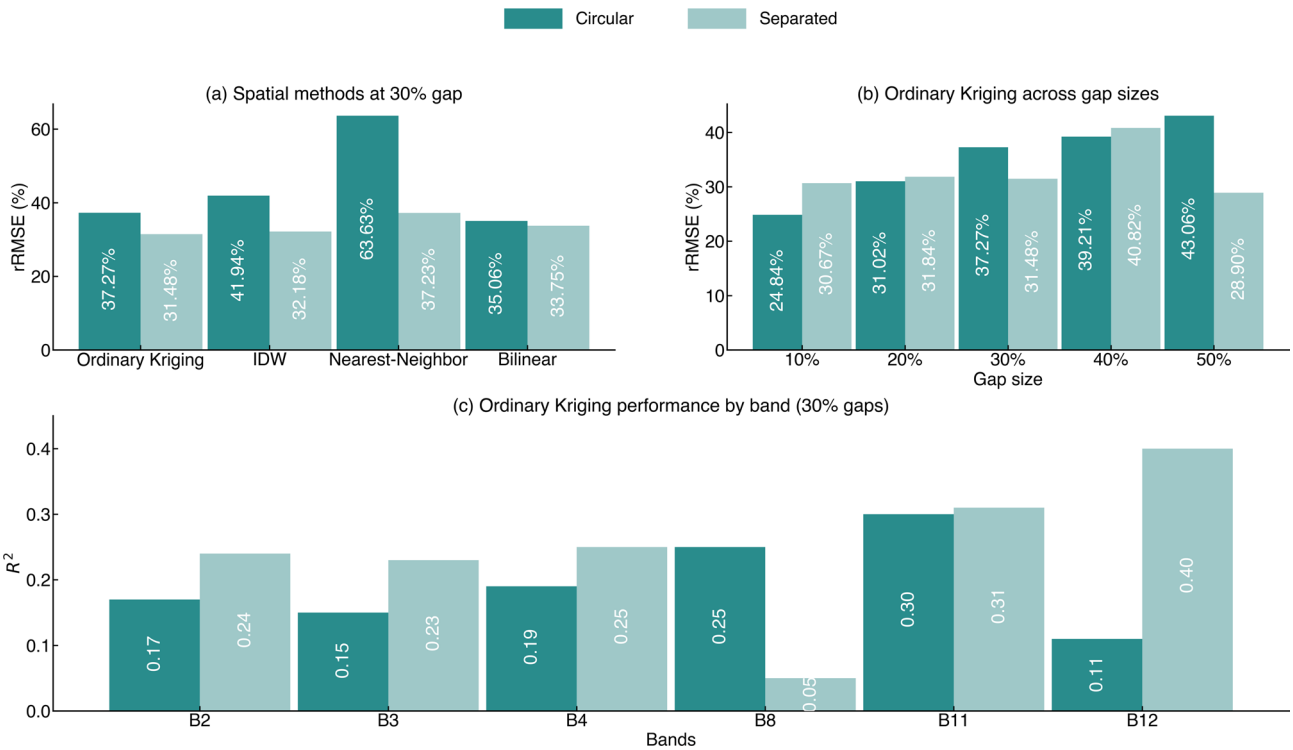


Fig. 11. Sentinel-2 reflectances gap filling using spatial methods: (a) Comparison of the performance of all spatial methods for 30% of circular and separated gaps, (b) Performance of the ordinary kriging method for the spatial gap size for circular and separated gap, (c) Performance of the ordinary kriging method for the different Sentinel-2 bands for circular and separated gaps.

Method	Size	Circular Gap			Separated Gap		
		rRMSE (%)	R^2	Bias	rRMSE (%)	R^2	Bias
Ordinary Kriging	10%	24.84	0.18	0.0014	30.67	0.25	0.0038
	20%	28.21	0.01	0.0018	31.84	0.15	0.0050
	30%	37.27	0.10	0.0047	31.48	0.23	0.0039
	40%	39.70	0.07	0.0035	40.81	0.13	0.0058
	50%	42.85	0.11	0.0041	28.89	0.21	0.0037
IDW	10%	32.24	0.03	0.0023	30.84	0.32	0.0038
	20%	39.96	0.09	0.0036	35.88	0.08	0.0064
	30%	41.94	0.09	0.0040	32.18	0.20	0.0040
	40%	43.44	0.08	0.0042	40.42	0.10	0.0057
	50%	46.58	0.09	0.0049	30.76	0.21	0.0042
Nearest-Neighbor	10%	33.60	0.05	0.0025	40.77	0.09	0.0067
	20%	50.32	0.01	0.0057	38.27	0.12	0.0072
	30%	63.63	0.02	0.0091	37.22	0.14	0.0054
	40%	60.80	0.01	0.0083	54.57	0.05	0.0104
	50%	74.82	0.02	0.0126	36.19	0.15	0.0058
Bilinear	10%	24.14	0.13	0.0013	41.32	0.01	0.0069
	20%	34.45	0.05	0.0027	35.24	0.08	0.0061
	30%	35.05	0.07	0.0028	33.75	0.18	0.0044
	40%	29.62	0.02	0.0020	39.31	0.15	0.0054
	50%	98.46	0.15	0.0218	32.05	0.15	0.0045
Bicubic	10%	146.18	0.003	0.0480	217.90	0.01	0.1904
	20%	510.56	0.04	0.5855	190.51	0.01	0.1793
	30%	599.55	0.01	0.8074	109.39	0.01	0.0466
	40%	339.02	0.001	0.2581	337.11	0.05	0.3959
	50%	4083.09	0.06	37.4451	166.88	0.01	0.1229

Table 4. Overall accuracy table of the Spatial gap filling methods across different Spatial gap type and size for band 4.

at 30% of sequential temporal gaps. However, ordinary kriging proved more stable and effective performance across larger gap sizes of 20%, 30%, 40%, and 50% of temporal separated gap in case of vegetation cover, with rRMSE values of 4.98%, 5.39%, 5.41%, and 5.49%, respectively (Fig. 12b), and hence slightly outperforming IDW (rRMSE of 4.94%, 5.82%, 6.05%, and 6.25% for the same type, size of gaps and cover type) (Table 5).

The performance of each method decreases gradually as the gap size increases in both types of temporal gap (Table 5). For instance, in case of vegetation cover type and the ordinary kriging, the rRMSE increased from 2.99% at 10% to 7.20% at 50% gap size for sequential gaps. For separated gaps, the rRMSE values increased from 3.94% at 10% to 5.49% at 50%. Similarly, the Nearest-Neighbor method showed an increase in rRMSE from 5.87% at 10% to 7.50% at 50% for sequential gaps, and from 6.34% at 10% to 10.69% at 50% for separated gaps. IDW showed an rRMSE increase from 2.88% at 10% to 6.50% at 50% for sequential gaps, and from 3.27% to 6.25% for separated gaps. In addition, the result shows that the temporal gap filling methods generally perform better for separated gaps (Fig. 12c). Cubic spline interpolation exhibited a significant mismatch, with rRMSE values surging from 9.58% to 34.67% for sequential gaps, but only increasing from 4.17% to 9.38% for separated gaps. Linear interpolation and IDW also followed this trend, with rRMSE values increasing more gradually for separated gaps.

The effect of land cover types on gap filling methods in Sentinel-2 bands is noticeable, with each cover type (vegetation cover, bare soil and water bodies) yielding different levels of interpolation accuracy. For small gaps, 10%, the ordinary kriging performs more effectively on bare bare soil with an rRMSE of 0.99% compared to water bodies, where it records a slightly higher rRMSE of 2.94%. As gaps expand to 50%, the performance of methods remains more consistent over vegetation cover and water bodies, maintaining lower rRMSE values. In contrast, over the bare soil, the interpolation accuracy decreases, reaching rRMSE values around 9.34% (Fig. 12b). On the other hand, using the ordinary kriging in the gap filling across Sentinel-2 spectral bands is generally consistent, particularly in the visible and NIR bands with high R^2 values in vegetation cover type (i.e., 0.84 for B02, 0.79 for B03, 0.79 for B04 and 0.97 for B08) (Fig. 12d). However, the SWIR bands show a more pronounced decline in interpolation accuracy with an R^2 values of about 0.94 in B11 and 0.91 in B12.

Spatio-Temporal results

The result of the spatio-temporal methods reveals a clear distinction in performance between the methods in the different gap scenarios (Fig. 13a). For spatial circular gaps, CLR reached an rRMSE value of 8.12% for all sizes. While for spatial separated gaps, the rRMSE values ranged from 5.05% at 10% to 5.85% at 50% gap size (Fig. 13b). This demonstrates its capability to maintain high performance, regardless of the type or size of the

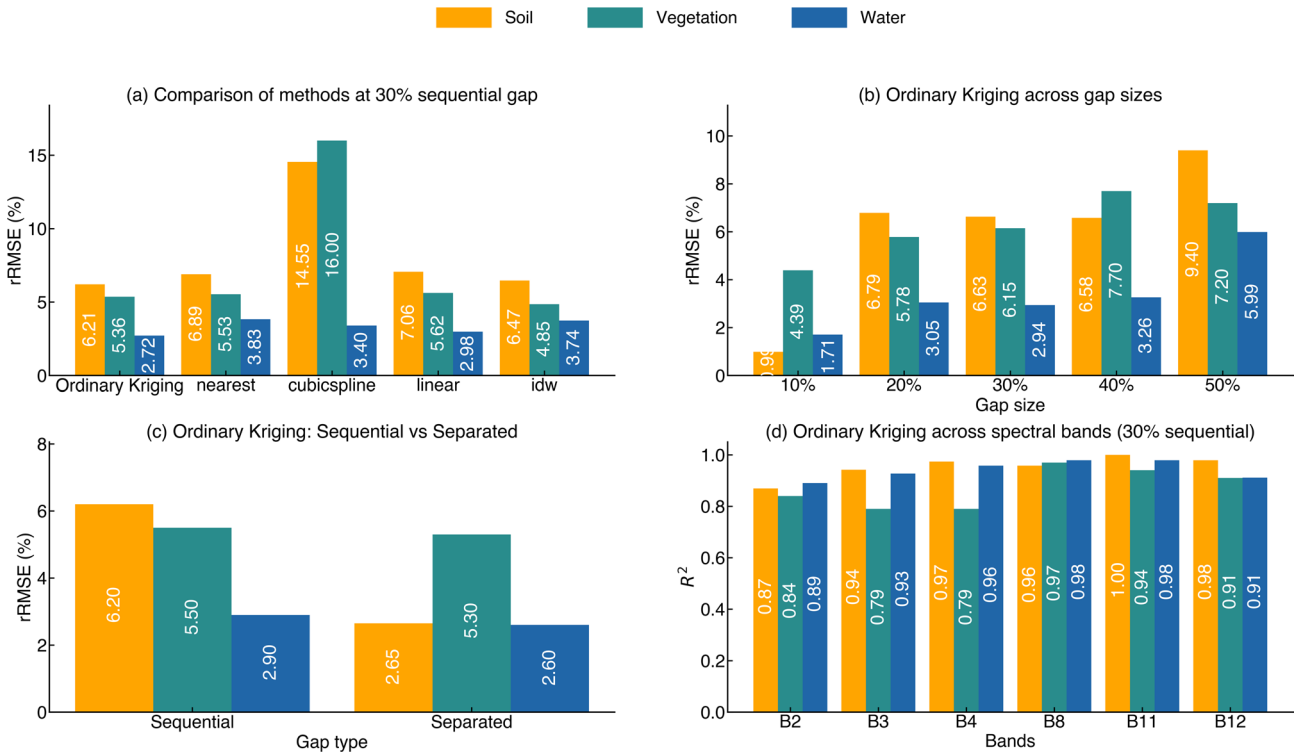


Fig. 12. Sentinel-2 reflectances gap filling using temporal methods: (**a**) Comparison of the methods performance for 30% of sequential temporal gaps, (**b**) Performance of the ordinary kriging method for temporal sequential gap by gap size and land cover type, (**c**) rRMSE (%) of the ordinary kriging method between sequential and separated temporal gaps, (**d**) ordinary kriging method performance by bands for 30% spatial and temporal gaps.

spatial gap. Similarly, CLR method demonstrated an ability to handle sequential temporal gaps with rRMSE values under 7.45% and less than 6.96% for separated temporal gaps for all sizes (Fig. 13c).

The DL method proved resilient performance in most cases similarly to the CLR method. However, it exhibited underperformance in scenarios involving temporal sequential gaps. In this case rRMSE ranged from 5.96% at 10% to 18.62% at 50% gap size, which is higher than all the other methods (Table 6). Despite this slight underperformance, DL exhibited strong adaptability in spatial gap scenarios, where its rRMSE for circular gaps is 11.74% at 10% size and marginally increased to 11.83% at 50% size. Similarly for spatial separated gaps, the rRMSE of DL method is always under 11.46%.

The performance of the DTWS method is generally below CLR and DL in most scenarios, except for the circular spatial gap where it is notably superior to DL and close to CLR. DTWS shows a decreasing performance gradient with increasing gap sizes in both spatial and temporal gaps. For instance, in temporal sequential gaps, DTWS performance showed an rRMSE of 12.90% at 10% size and increased to 14.96% at 50% size, reflecting a trend of decreasing accuracy with larger temporal discontinuities (Table 6). However, in case of a spatial circular gap, the DTWS method shows better results compared to DL method with a low rRMSE value of 10.75% at 30% compared to 11.83% at same type and size of gap.

The performance of the CLR method for different Sentinel-2 bands demonstrates a high accuracy, particularly in the visible spectrum bands. The CLR method achieves an R^2 values of 0.96 in the Blue (B02), 0.97 in the Green (B03), and 0.97 in the Red (B04) bands in spatial separated gaps and 0.82, 0.86, 0.89, respectively, in spatial circular gap (Fig. 13d). In the Near-Infrared (B08) band CLR method attained 0.92 in spatial circular gap and 0.94 in separated gaps. Similarly, in the Short-Wave Infrared bands (B11 and B12) this method demonstrates robustness with an R^2 values of about 0.85 and 0.96 for B11 and B12, respectively, in the spatial circular gap case and 0.93 and 0.97 in the spatial separated gap (Fig. 13d).

Spatio-Spectral results

Figure 14; Table 7 illustrate the results of the spatial-spectral random forests (SSRF) method for the different tests performed in this study. The evaluation results of the band combinations, used in the training of SSRF to fill the gap in B04 showed that the combination of bands B02, B03, and B08 is the most accurate in terms of metrics (rRMSE = 8.9% / 9.18%, R^2 = 0.87 / 0.93 and bias = 0.0001 / 0.0003 for circular gaps and separated gaps, respectively). In addition, the other bands combinations (B02), (B03), (B02, B03), (B02, B08) and (B03, B08) also proved a competitive performance to the combination of (B2,B3,B8) with rRMSE values of 10.3%, 12.18%, 9%, 10.03% and 10.02%, respectively, for circular gap, and 13.01%, 11.53%, 10.13%, 9.84% and 9.44% for separated gaps (Fig. 14a).

method	size	Sequential Gaps			Separated Gaps		
		rRMSE (%)	R^2	Bias	rRMSE (%)	R^2	Bias
Ordinary Kriging	10%	2.99	0.94	-0.001	3.94	0.9	-0.0002
	20%	5.26	0.81	0.0002	4.98	0.83	0.0007
	30%	5.44	0.79	-0.0006	5.39	0.8	9.23×10^{-5}
	40%	5.75	0.77	-0.0009	5.41	0.8	-0.0007
	50%	7.20	0.66	0.001	5.49	0.79	-0.0009
Nearest-Neighbor	10%	5.87	0.61	-0.001	6.34	0.89	0.0007
	20%	5.58	0.61	0.00075	7.38	0.82	0.0015
	30%	5.63	0.23	0.0002	9.16	0.61	-0.0003
	40%	9.80	0.31	-0.005	9.20	0.58	-0.0006
	50%	7.50	0.0	0.004	10.69	0.55	-0.002
Cubic spline interpolation	10%	9.58	0.95	-0.004	4.17	0.93	-4.2×10^{-6}
	20%	7.86	0.84	-0.002	5.21	0.83	0.0005
	30%	15.95	0.83	0.011	8.24	0.77	-0.0008
	40%	23.23	0.79	-0.02	8.6	0.76	-0.001
	50%	34.67	0.74	0.03	9.38	0.75	-0.003
linear interpolation	10%	5.87	0.81	-0.002	4.64	0.87	0.0001
	20%	5.89	0.76	0.0006	5.99	0.76	0.001
	30%	5.70	0.77	0.0002	6.95	0.69	-0.0005
	40%	7.68	0.7	-0.005	7.15	0.68	-0.0009
	50%	8.39	0.58	0.004	8.00	0.64	-0.0023
IDW	10%	4.90	0.84	-0.001	3.27	0.79	-8.66×10^{-5}
	20%	4.93	0.78	0.0009	4.94	0.68	0.0008
	30%	4.90	0.78	-0.0002	5.82	0.57	-0.0006
	40%	25.61	0.62	-0.001	6.05	0.57	-0.001
	50%	6.5	0.69	0.002	6.25	0.51	-0.0014

Table 5. Overall accuracy of the Temporal gap filling methods across different Temporal gap type and size for the band 4 and vegetation land type.

The performance of the SSRF method is evaluated against the effect of period duration between the cloudy and known images. This test shows that prediction accuracy is affected by temporal factors. The images acquired five days before (31 May 2022) and five days after (10 June 2022) the cloudy image (5 June 2022) showed the best performance. The result shows an rRMSE values of 8.94% (31 May 2022) and 8.90% (10 June 2022) in circular gap and 8.66% (31 May 2022) and 9.18% (10 June 2022) in separated spatial gaps (Fig. 14b). These values are comparatively lower than 9.29% (11 May 2022) and 11.32% (30 June 2022) in the circular gap, and 10.97% and 12.25% in the separated gap, obtained for the images that distant 10 days (11 May 2022, 30 June 2022) from the cloudy image (Fig. 14b).

The SSRF method exhibits strong performance for different types and sizes of the spatial gap. For circular gaps, rRMSE values vary from 7.38% at 10% to 8.30% at 50% gap size with high R^2 values above 0.87, indicating high level of prediction accuracy (Table 7). Similarly, the SSRF method showed high efficacy in separated gaps with rRMSE value of 6.03% at 10% gap size and an R^2 of 0.97 (Table 7). This method affirms good predictive performance that slightly decreases to an rRMSE value of 6.95% at 50% gap size and an R^2 of 0.95. The R^2 values for SSRF are uniformly high across all Sentinel-2 bands with R^2 values of 0.95 and above, except the B11 where the method results a R^2 value of 0.78 in circular gap and 0.77 in separated gap (Fig. 14c).

Visual evaluation

The visual quality of the gap filled images is evaluated by focusing on the spatial continuity, blurring, noise, and unnatural transitions. Spatial methods generally give the worst results among the different categories of the evaluated methods (Fig. 15), and within this category (spatial), ordinary kriging and IDW are relatively the best compared to other spatial methods, with SSIM values of about 0.7198 and 0.7163, respectively (Table 8). Ordinary kriging is effective in creating smooth transitions in areas with gradual changes in reflectance, although it can introduce slight blurring as the image is divided into four parts before filling the gaps. Similarly, the IDW also maintains good spatial continuity in homogeneous areas but may introduce noise and unnatural transitions in regions with sudden changes. In contrast, the Nearest-Neighbor method achieves an SSIM of about 0.6593 and fails to maintain smooth transitions, yielding a less natural appearance. Linear interpolation performs moderately well with good spatial continuity and minimal noise (Fig. 15) and achieves an SSIM of 0.7068. Nevertheless, it exhibits some blurring in areas with gradual transitions. Notably, the cubic spline interpolation method performs poorly compared to other spatial methods achieving an SSIM of about 0.2975. This method introduces significant blurring and highly unnatural transitions, visible as distorted regions.

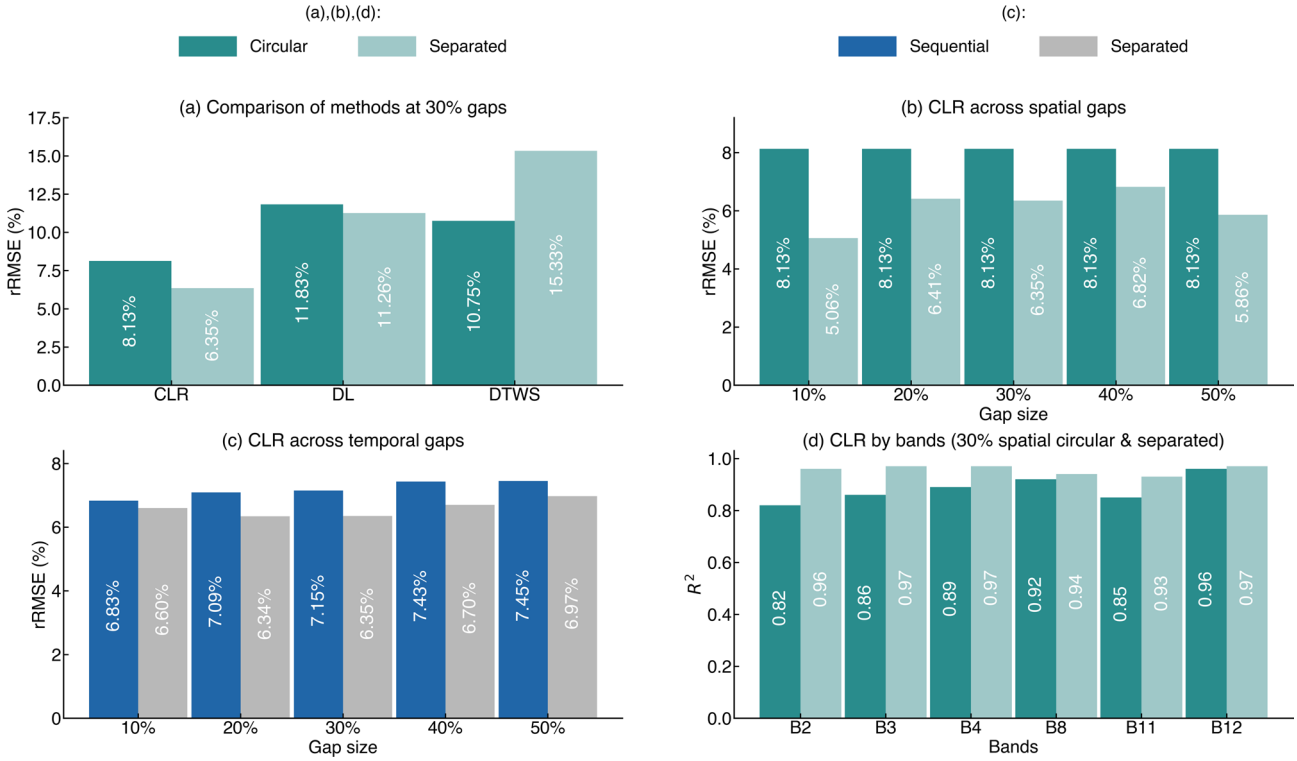


Fig. 13. Sentinel-2 reflectances gap filling using spatio-temporal methods for the spatial and temporal gaps: (**a**) Performance comparison for 30% spatial and temporal gaps, (**b**) Performance of CLR method for different sizes of the spatial circular and the separated gaps, (**c**) Performance of CLR method for different sizes of the temporal sequential and the separated gaps, (**d**) Performance of CLR method by Sentinel-2 bands for 30% of the spatial circular and the separated gaps.

Gap type		CLR			DL			DTWS		
	Gap size	rRMSE (%)	R^2	Bias	rRMSE (%)	R^2	Bias	rRMSE (%)	R^2	Bias
Spatial Circular	10%	8.12	0.89	0.0001	11.74	0.77	-0.012	11.90	0.76	-0.004
	20%	8.12	0.89	0.0001	11.83	0.77	-0.013	10.47	0.82	-0.006
	30%	8.12	0.89	0.0001	11.83	0.77	-0.013	10.75	0.81	-0.007
	40%	8.12	0.89	0.0001	11.83	0.77	-0.013	10.88	0.80	-0.008
	50%	8.12	0.89	0.0001	11.83	0.77	-0.013	10.98	0.80	-0.008
Spatial separated	10%	5.05	0.97	0.0001	10.798	0.90	-0.016	10.11	0.91	-0.002
	20%	6.41	0.96	0.0002	11.26	0.89	-0.018	12.53	0.86	-0.012
	30%	6.35	0.97	0.0001	11.26	0.89	-0.016	15.33	0.81	-0.009
	40%	6.82	0.97	0.0001	11.46	0.92	-0.015	11.98	0.91	-0.006
	50%	5.85	0.97	0.0001	10.72	0.89	-0.017	9.93	0.90	-0.008
Temporal sequential	10%	6.83	0.96	0.0001	5.96	0.97	-0.0006	12.90	0.87	-0.004
	20%	7.08	0.96	0.0001	11.34	0.89	0.0077	13.57	0.85	-0.006
	30%	7.15	0.95	0.0001	14.42	0.92	0.0100	15.84	0.80	-0.012
	40%	7.43	0.95	0.0002	18.22	0.89	0.0143	14.60	0.83	-0.011
	50%	7.45	0.95	0.0002	18.62	0.90	-0.0197	14.96	0.82	-0.010
Temporal separated	10%	6.60	0.96	0.0001	6.39	0.96	-0.0001	14.52	0.83	-0.006
	20%	6.34	0.96	0.0001	6.26	0.96	-0.0006	15.46	0.80	-0.007
	30%	6.35	0.96	0.0001	7.42	0.95	0.0019	15.33	0.81	-0.009
	40%	6.69	0.96	0.0001	7.76	0.95	-0.005	14.97	0.82	-0.008
	50%	6.96	0.96	0.0001	9.50	0.92	0.006	14.75	0.82	-0.006

Table 6. Statistical metrics of the spatio-temporal gap filling methods for the different gap type and size.



Fig. 14. Sentinel-2 reflectances gap filling using the spatial-spectral random forests (SSRF) method: (a) Performance of SSRF by bands combinations used in training, (b) Performance of the SSRF method using different dates of known image, (c) The performance of SSRF method across different bands.

	Circular Gap			Separated Gap		
Size	rRMSE (%)	R^2	Bias	rRMSE (%)	R^2	Bias
10%	7.38	0.91	0.0001	6.03	0.97	0.0001
20%	8.5	0.88	0.0001	8.29	0.94	0.0003
30%	8.9	0.87	0.0001	9.18	0.93	0.0003
40%	8.25	0.88	0.0001	8.57	0.95	0.0002
50%	8.30	0.88	0.0001	6.95	0.95	0.0002

Table 7. Statistical metrics computed using the spatial-spectral random forests (SSRF) method in gap filling of B04 using image dated 10 June 2022 as known image and (B02, B03, B08) bands combination for different gap type and size.

Spatio-temporal methods are extremely efficient for maintaining spatial continuity and reducing visual distortions. DTWS method maintains very good spatial continuity, preserving the natural structure of the landscape, with only slight blurring and minimal noise (Fig. 15), and reaches an SSIM of 0.9429 (Table 8). Unnatural transitions are rare, resulting in generally smooth transitions. The spatio-temporal DL and CLR methods provide excellent spatial continuity and minimal blurring (Fig. 15), with SSIM values of about 0.9840 and 0.9822, respectively. They also exhibit low noise levels and natural looking transitions with very few noticeable artifacts.

The SSRF method stands out as the best performing method among all the tested gap filling methods in terms of visual evaluation. It offers outstanding spatial continuity, capturing complex patterns with high accuracy. SSRF almost completely avoids blurring, preserving fine details exceptionally well and maintaining extremely low noise levels (Fig. 15), and it delivers the highest SSIM among all methods at 0.9856 (Table 8). The transitions are natural looking, with negligible artifacts, providing the highest visual quality among the methods evaluated.

Running time comparison

Assessing the efficiency of various gap filling methods involves comparing their running times on filling the gap within a single image. These methods are executed on an Intel Xeon Gold 6152 CPU with 22 cores and 44 threads, and 128 GB of RAM. All algorithms are implemented using Python version 3.8.13 under the Anaconda distribution (anaconda/3-2022.05) on a Linux system. There are significant differences in computational times among the tested methods. The interpolation methods (Nearest-Neighbor, Bilinear, Bicubic) consumed an execution time less than 0.01 min followed by the IDW method that required 2.98 min (Table 9). However,

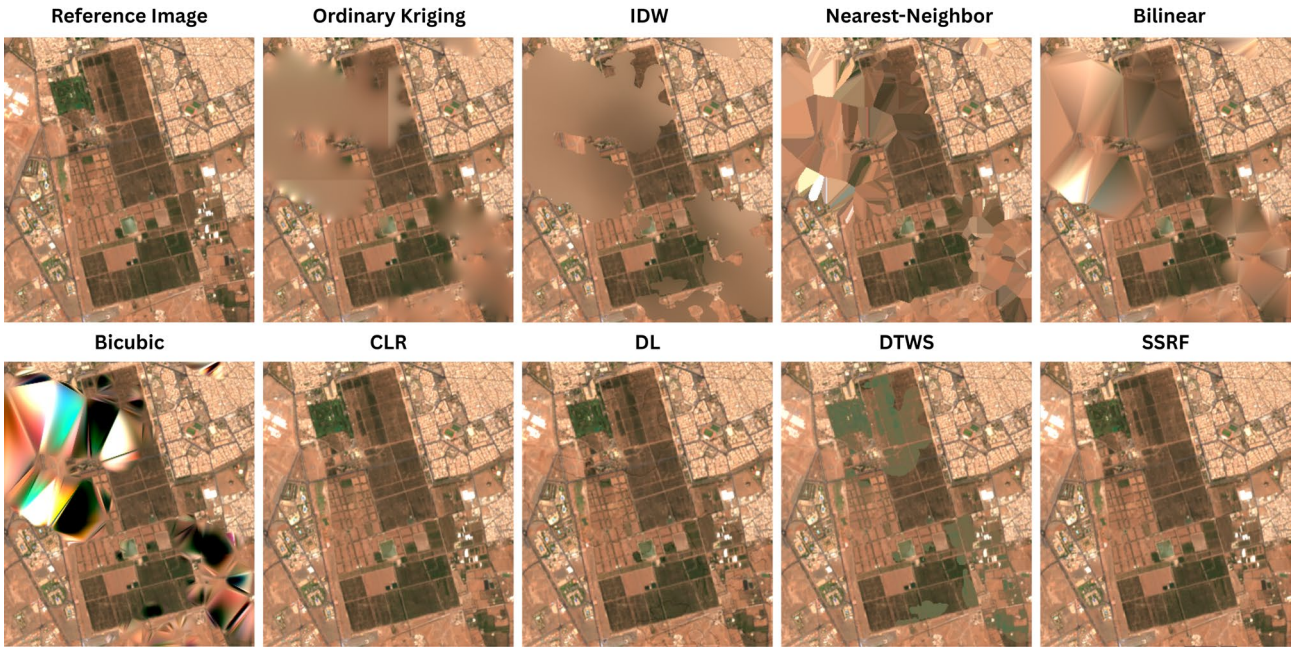


Fig. 15. Visual comparison of gap-filling methods in Sentinel-2 RGB (natural color, bands B04/B03/B02) images over the study area. All panels are derived from Sentinel-2 Level-2 data obtained via Google Earth Engine, with RGB composites generated in Python using Matplotlib 3.10[67].

Method	SSIM
Ordinary Kriging	0.7198
Inverse Distance Weighting (IDW)	0.7163
Nearest-Neighbor	0.6593
Bilinear	0.7068
Bicubic	0.2975
Clustering and Linear Regression (CLR)	0.9822
Dynamic Time Warping Segmentation (DTWS)	0.9429
Deep Learning (DL)	0.9840
Spatial-Spectral Random Forests (SSRF)	0.9856

Table 8. Comparative SSIM results for Gap-Filled Sentinel-2 RGB composites (natural color) compared to Ground-Truth Reference.

Method	Running Time (minutes)
Ordinary Kriging	96.76
Inverse Distance Weighting (IDW)	2.98
Nearest-Neighbor	0.003
Bilinear	0.01
Bicubic	0.02
Clustering and Linear Regression (CLR)	2.16
Dynamic Time Warping Segmentation (DTWS)	4.83
Deep Learning (DL)	Prediction: 0.11, Training: 180
Spatial-Spectral Random Forests (SSRF)	6.84

Table 9. Execution time comparison of various gap filling methods.

ordinary kriging exhibits a considerably longer execution time of 96.76 min. The spatio-temporal methods DTWS and CLR took 4.83 and 2.16 min, respectively, followed by 6.84 min for the SSRF method (Table 9). Notably, the Deep Learning method stands out for its rapid prediction time of 0.11 min. Yet, it requires a significantly longer training phase, taking approximately three hours on an NVIDIA A100 GPU with 24 GB of VRAM.

Discussion

This study comprehensively evaluated the performance of various gap filling methods for Sentinel-2 imagery across different gap types, sizes, and spectral bands. The key findings reveal significant differences in the effectiveness and computational efficiency of each method, highlighting their advantages and limitations in various scenarios.

Spatial gap-filling methods

For spatial methods, the results indicate different levels of efficacy between circular and separated gap types. This suggests that geostatistical and interpolation methods, which depend strongly on the spatial autocorrelation between data points, are better suited to gaps where this correlation is more predictable and uniform. This is typically the case in small sizes of circular gaps. Conversely, the separated gaps, which are more irregular and dispersed, likely present challenges in spatial continuity. These challenges disrupt the assumptions underlying geostatistical and interpolation methods. The results clearly demonstrate (Figure A- 1) that as the gap sizes increase, all the spatial methods exhibit a trend of suboptimal performance. This trend is related to the reduced availability of nearby valid data points in larger gaps. As a result, the data required for interpolation is taken from further pixels. Indeed, this increases the likelihood of introducing error due to less similarity or homogeneity between the data points. In addition, the high spatial resolution of Sentinel-2 images significantly affects the performance of gap filling methods. For instance, a 10% gap size in an image of 364×320 pixels represents a substantial area. It also introduces complexities in interpolation due to the increased amount of detailed information missing. This gap, amounting to over 13,000 pixels, challenges the methods significantly as they must interpolate across a broader spatial extent with variable surface characteristics. Moreover, the geostatistical methods are too slow, especially the ordinary kriging, which is most performant in terms of accuracy among the spatial methods. This slow processing time becomes particularly problematic at large-scale or time-sensitive applications where rapid data processing is crucial. The previous studies such as[18] have also highlighted the slow speed of geostatistical methods as a primary disadvantage and also their usefulness in filling small spatial gaps where high spatial autocorrelation exists. Additionally, it has been shown that ordinary kriging may perform poorly in complex air temperature mapping[41].

Temporal gap-filling methods

Regarding temporal methods, the results highlight a distinct performance pattern between sequential and separated gaps. Generally, both ordinary kriging and IDW performed better with smaller separated gaps, exploiting the temporal continuity present in the datasets. This observation is consistent with the findings of[42] demonstrated that gap-filling methods such as gaussian process regression and spline interpolation perform better when temporal continuity is preserved. However, as gap size increased, particularly to 50%, the performance of all methods, including linear and cubic spline interpolations, tends to degrade. This is most pronounced with cubic spline interpolation, where rRMSE values raised to 34.67% in sequential gaps, suggesting substantial challenges in bridging larger temporal discontinuities. In separated gaps, the performance degradation is less severe, as also noted by[42], suggesting that these methods are better suited to handling datasets where missing data points are separated rather than sequential.

The analysis underscores the variable performance of ordinary kriging in interpolating temporal gaps for different land cover types (Figure A-2). Each type exhibits distinct temporal patterns that impact the method effectiveness. For water bodies, there is a strong alignment between observed and predicted values, showcasing ordinary kriging proficiency in environments with stable temporal patterns. These results are seen in the study[39] where stable land cover types exhibited more reliable interpolation results. Similarly, the performance of ordinary kriging in vegetation cover type shows a general reliability in filling the gaps, which can be related to the slow phenological changes as in olive trees. This reliability of performance is likely linked to missing peaks in the time series of reflectance during the study period. In contrast, the bare soil cover type reveals a more volatile reflectance pattern with sharp and erratic peaks that ordinary kriging struggles to predict accurately. These peaks are likely due to moisture changes because of rainfall/irrigated events. This is similar to the findings from[39] regarding the challenges that interpolation methods face in dynamic environments. This highlights the limitations of ordinary kriging in capturing rapid temporal variations. Although ordinary kriging performs well in stable conditions like water and arboriculture, its application in more dynamic environments, such as bare soil and seasonal agriculture with high phenological changes, requires careful consideration of each cover type specific temporal dynamics to enhance prediction accuracy.

Spatio-Temporal methods

The evaluation result of the CLR method consistently demonstrated high accuracy across different scenarios. Leveraging spatial and temporal correlations, CLR effectively fills the gaps regardless of the size or the type of gaps. This is evident from the tight clustering of data points along the diagonal line of perfect prediction (Figure A- 3). The methodology of CLR ensures robust prediction by focusing on the representative features that are sampled from the selected similar regions rather than isolated pixels that have less temporal representativity. This result aligns with those obtained by[50] that also utilized clustering and regression-based strategies for gap filling on LST data derived from Landsat-8.

The DL method also maintains strong performance across different gap sizes and types. The slightly larger spread of data points around the diagonal line (Figure A- 3), especially in larger gaps, suggests a minor underestimation bias. This becomes more pronounced as gap size increases. In circular spatial gaps, the DL method leverages its advanced pattern detection capabilities to interpolate data efficiently. This is due to its ability to learn complex and non-linear relationships within the data. Which is ideal for predicting values in uniform gap distributions where contextual information is the key. However, DL method performance can sometimes be computationally intensive, requiring significant training time and resources, which can be a disadvantage in resource-limited conditions. For separated spatial gaps, the DL method continues to perform robustly, utilizing learned spatial correlations from extensive training on diverse datasets. Its ability to process large amounts of data and learn from them allows DL to handle these irregular and dispersed gaps with high accuracy. Nevertheless, the training process can be sensitive to the quality and variety of training data. This makes it vulnerable to overfit when the training data is not representative enough to the scenarios encountered during the period of interest. In addition, the DL approach shows significant underperformance in sequential temporal gaps, which can be linked to the limited temporal context provided by the small block size of only eight images. This limits the model ability to access extended sequences of data, which is a key factor for understanding temporal dynamics and making accurate predictions. When gaps span multiple images or a substantial portion of these blocks, the model struggles due to insufficient data to effectively interpolate missing values. This issue is further aggravated by the model challenges in handling discontinuities within the data. In contrast, in separated temporal gaps, the DL method advanced feature learning enables it to interpolate discontinuous time points effectively and predict values even with irregular intervals between observations. This makes DL particularly useful in scenarios like cloud-covered satellite data where observations are intermittently missing. The effectiveness of DL in these cases heavily depends on the initial model architecture and the extent of tuning. This can require extensive experimentation and adjustment, thus increasing the complexity of model deployment. This result, like the one founded by[52], demonstrates the efficacy of deep learning models with partial convolutions for gap filling in satellite imagery, achieving fast computation times and comparable accuracy to other methods. Additionally, the study[53] proposed the Content-Sequence-Texture Generation (CSTG) network, which uses deep neural networks to fill the gaps in MODIS data. The CSTG method has proven high accuracy, especially with large gaps. This demonstrates the effectiveness of deep learning architecture in the context of gap filling despite the difficulties in the training and computational process.

The DTWS method shows a varied performance based on the nature and complexity of the gaps. It performs notably well in circular spatial gaps reflecting its ability to exploit uniformity and continuity within these spatial conditions. However, DTWS struggles with more challenging gap scenarios, such as separated spatial gaps and sequential temporal gaps, where its performance declines as gap sizes grows. This trend underscores DTWS reliance on data continuity, which becomes a limitation in handling complex and discontinuous gaps.

Spatio-Spectral methods

The evaluation results of the SSRF method for gap filling yield valuable insights. The combination of bands B02, B03, and B08 consistently achieves the lowest rRMSE in gap filling of the B04 band comparing to the other combination used during training phase of the SSRF. The combinations of one or two bands show slightly higher errors. This indicates that SSRF effectively leverages spectral information even with fewer bands used in the training. The exception is the combination that contains only the B08 band, which shows a remarkably high rRMSE. This finding demonstrates that the SSRF method can be adaptable to available data, balancing predictive accuracy and data availability. Another important parameter for the performance of the SSRF method is the temporal proximity of the known image to the target image. This directly influences the model predictive accuracy. Using a known image in training that is closer to the date of the target image significantly improves SSRF predictions. This minimizes prediction errors caused by seasonal variations and land cover changes. Moving to the effectiveness of this method by gap type and size, the SSRF maintained high efficacy across both circular and separated gaps. This adaptability is also evident as gap sizes increased from 10% to 50%, with rRMSE values remaining below 9%. The SSRF robustness against increasing gap sizes and complexity reveals its ability to generalize and fill gaps effectively. This result aligns with the finding of the study[21] that demonstrated the efficacy of SSRF method in gap filling of Landsat 8 and Sentinel-2 data. Additionally, the recent work[33] demonstrated that the STRF method excels in addressing large gaps and significant temporal changes compared to traditional methods. The main difference between SSRF and STRF is that STRF incorporates multi-scale data from both spatially complete Landsat and GOES LST images (fusion), where SSRF leverages multispectral information from a single sensor. Both methods enhance accuracy by leveraging spectral information and random forest models, but SSRF shows efficiency and adaptability in various scenarios, especially when fewer spectral bands are available.

Advantages and limitations of gap filling methods

Analyzing the various gap-filling methods reveals that each method has its distinct limitations and advantages. Starting with geostatistical and interpolation methods, particularly ordinary kriging. These methods are computationally expensive and slow. They struggle to handle larger, more complex gaps. They excel in filling smaller homogenous spatial gaps due to the high spatial autocorrelation present. Cubic spline interpolation is prone to extremely high errors when managing gaps. In contrast, ordinary kriging maintains a degree of accuracy with simpler and uniform patterns. Geostatistical and interpolation temporal methods also face challenges with larger temporal gaps, where they struggle to interpolate efficiently. Cubic spline interpolation tends to produce high errors with large sequential gaps, as it cannot handle rapid changes in temporal patterns. However, ordinary kriging and IDW perform consistently in small temporal gaps, while linear interpolation and Nearest-Neighbor effectively estimate values in moderately sized gaps. Spatio-temporal methods like CLR, DTWS, and

DL provide comprehensive solutions by combining spatial and temporal correlations. However, the DL method risks overfitting and requires significant computational resources, while DTWS accuracy declines with separated spatial and sequential temporal gaps. On the other hand, CLR consistently delivers robust performance across all gap types and sizes, leveraging both spatial and temporal dimensions. Finally, the SSRF method effectively balances spectral bands with spatial information. Despite requiring high quality training data and suitable band selection, it remains highly adaptable to varying data availability. It maintains high accuracy across spectral bands while effectively filling gaps.

Overall, although each technique excels in specific scenarios, CLR showed the most consistent performance across tested scenarios, but its applicability to other datasets and sensors still requires further validation. The SSRF method performed strongly, especially when suitable band combinations and temporally close reference images were available, though its accuracy declines with less favorable conditions. The DL method performed well in many spatial gap scenarios but struggled with long sequential temporal gaps and required extensive computational resources for training. Finally, the DTWS method performs adequately and can be improved by replacing the pixel weighting process with machine learning techniques to better align predicted values and enhance its performance across various scenarios. These categories of methods, especially when tailored to specific challenges, provide a comprehensive framework for gap filling in Sentinel-2 imagery.

Conclusion

In this study, various gap-filling methods are evaluated to address the issue of data gaps in Sentinel-2 imagery caused by cloud cover. The evaluation framework encompassed specific test scenarios for each category of methods, including spatial, temporal, spatio-temporal, and spatio-spectral methods. Additionally, running time comparisons and visual comparisons are conducted to assess the effectiveness and efficiency of each method. The findings of this study revealed that traditional spatial methods, such as ordinary kriging, provide suboptimal performance and are computationally expensive. This makes this category of methods useful only for small gaps with moderate performance. In contrast, advanced techniques like machine learning and deep learning models in spatio-temporal and spatio-spectral methods demonstrated superior performance in handling complex gap scenarios. The evaluation shows that each method has its limitations and specific scenarios where it performs better. Overall, CLR achieved the most consistent results across different gap types and sizes within our study period, though further work is needed to assess its robustness under real cloud conditions. Additionally, the results highlight the importance of selecting the appropriate gap filling method based on the specific requirements of the application. In conclusion, This study highlights the potential of machine learning and deep learning methods for gap filling in Sentinel-2 data, while also emphasizing their limitations and the need for scalable, sensor-independent approaches. Future studies should explore more generalizable and sensor-independent methods to refine these techniques further. They should also focus on the real-time application and scalability of these methods, especially considering the increasing volume of remote sensing data.

Data availability

All data and code utilized in this study are publicly available on GitHub. The repository can be accessed at the following link: https://github.com/said-grich/gap-filling. This repository contains detailed documentation and scripts required for data processing and analysis.

Appendixs

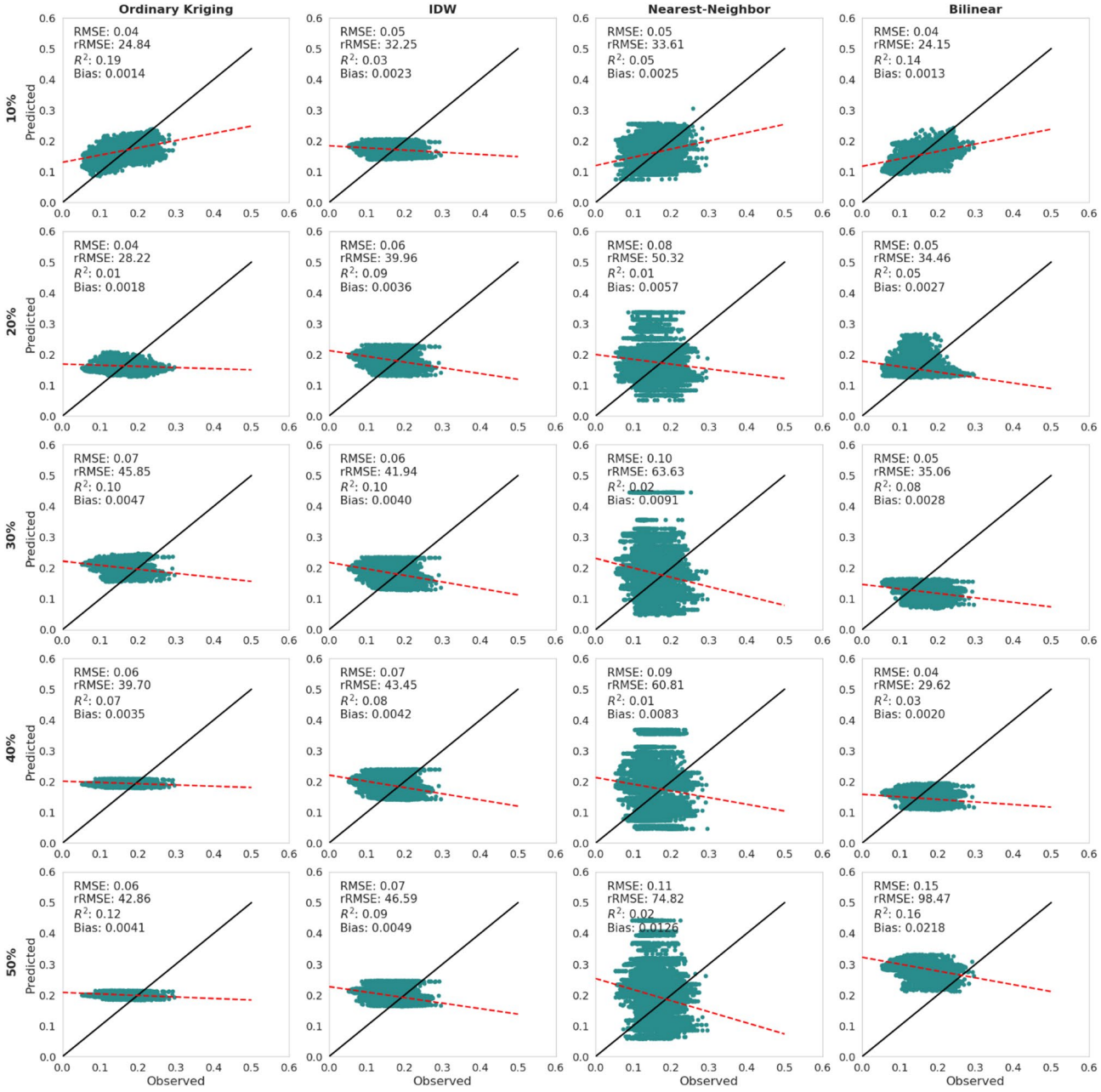


Fig. 16. Scatter plot of Predicted vs. Observed for spatial methods for band 4 and 30% of circular gap.

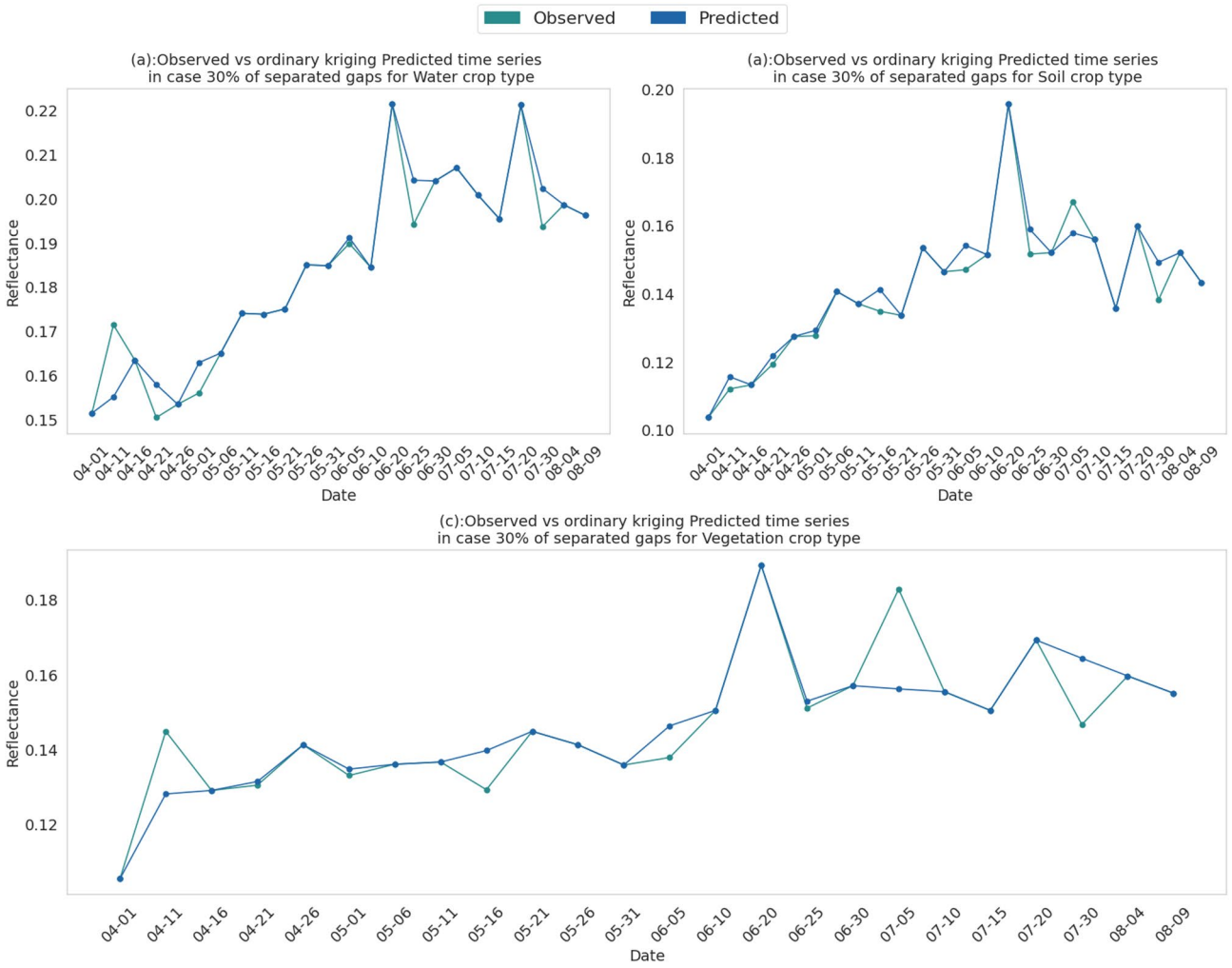


Fig. 17. Observed vs. Predicted of ordinary kriging method across different land cover types in case of 30% of temporal separated gap.

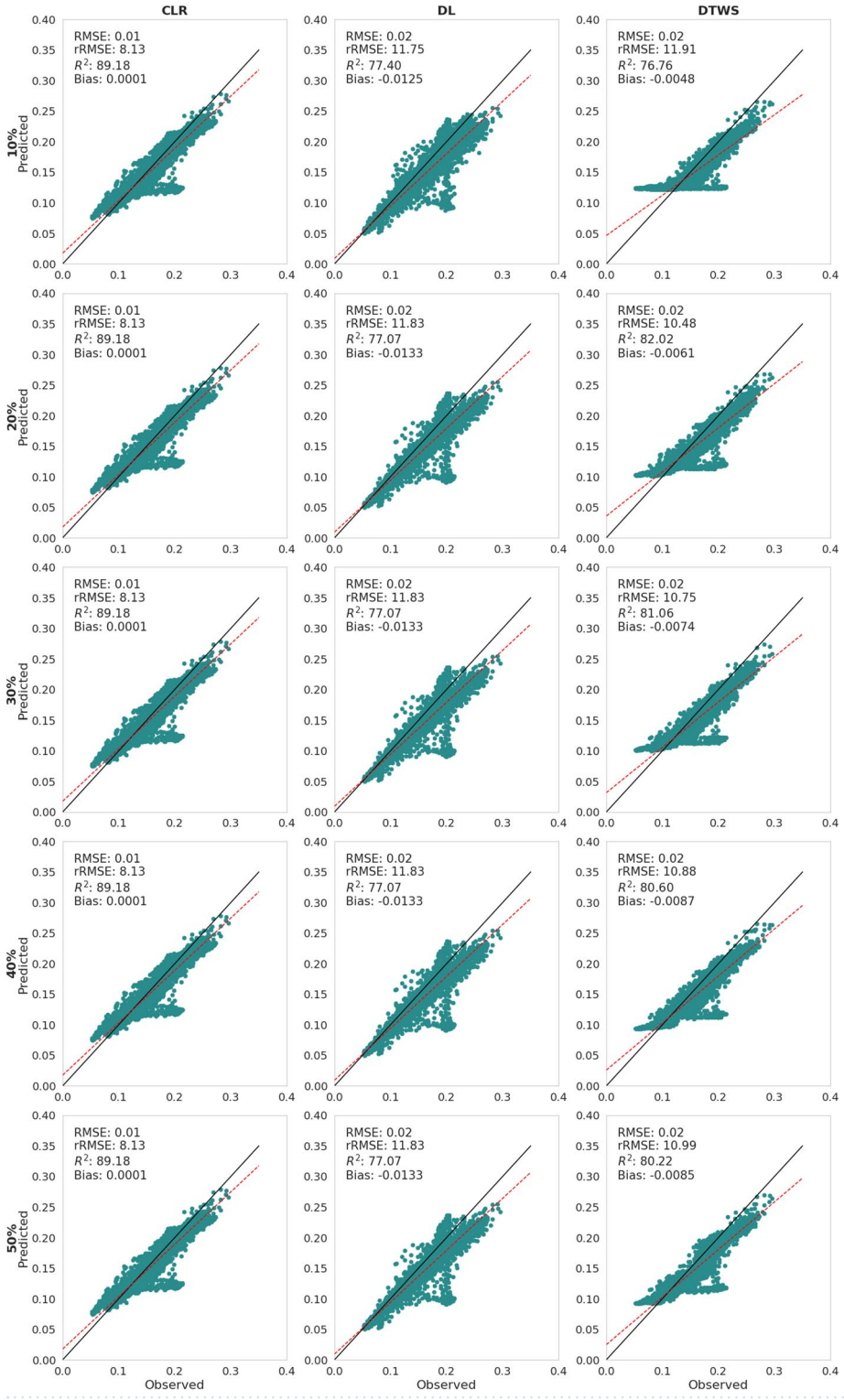


Fig. 18. Scatter plot of Predicted vs. Observed for spatio-temporal methods for band 4 and 30% of circular gap and 30% temporal separated gap.

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Author contributions

S.G. contributed to the development of the methodology, developed the algorithms, performed the data analysis, and wrote the initial draft of the manuscript. J.E., N.O., and B.A.H. proposed the initial idea for the study, contributed to the design of the research plan and objectives, and developed the methodology. H.H. assisted with algorithm development and data processing. J.E., N.O., and B.A.H. also assisted in interpreting the results and provided critical feedback on the manuscript. A.C. supervised the study and reviewed the final manuscript. All authors reviewed and approved the final version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Disclosure statement
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